

The barrier properties of sustainable multiphase and multicomponent packaging materials: A review

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ABSTRACT

The current packaging materials market is dominated by petrochemical-derived, single-use, and inexpensive plastics that are incredibly stable in the environment. After their sole use, only a small fraction is recycled, with the remaining ending up in landfills, incineration facilities, or littered in the environment threatening the health of our environment. Nature-derived, compostable, and biodegradable polymers are appealing alternatives to replace single-use plastics in packaging and coating applications. However, most sustainable polymer alternatives cannot compete with traditional plastics in key packaging property attributes, such as gas and moisture barrier properties. Thus, there is a dire need to design chemistries and processing technology to enhance the barrier properties of sustainable polymer systems. In this review, we conducted a critical and comprehensive summary of the important and impactful contributions on improving the gas and moisture barrier properties of sustainable polymer systems for packaging applications. The work highlights signature accomplishments on the effect of tuning the microstructure, surface energy, morphology, and self-assembly of polymers on the gas and moisture barrier properties of the resulting films and coatings. The recent developments in designing multiphase and multicomponent systems and the impact of nanotechnology on barrier properties of sustainable packaging plastics are emphasized.

1. Introduction

Polymer coatings and films, which act as stable decorative or protective layers of consumer goods, have been widely utilized in various engineering applications and rapidly growing over the past few decades [1,2]. Due to the rising popularity of take-out food in

Abbreviations: AC, Activated carbon; BNC, Bacterial nanocrystal; BW, Beeswax; CMC, Cellulose microcrystals; CNC, Cellulose nanocrystal; CNF, Cellulose nanofibril; CNT, Carbon nanotube; DCMC, Dialdehyde carboxymethyl cellulose; GO, Graphene oxide; GOCNT, Graphene oxide carbon nanotube; GNP, Graphene nanoplatelets; HCl, Hydrochloric acid; KBr, Potassium bromide; MA, Maleic anhydride; MAP, Modified atmosphere packaging; MMT, Montmorillonite; MTPS, Maleated thermoplastic starch; LNP, Lignin nanoparticles; NaOH, Sodium hydroxide; OMMT, Organo-Montmorillonite; OTR, Oxygen transmission rate; OP, Oxygen permeability; PBAT, Poly(butylene adipate terephthalate); PBS, Poly(butylene succinate); PCL, Polycaprolactone; PE, Poly(ethylene); PET, Poly(ethylene terephthalate); PGA, Poly(glycolic acid); PH, Poly(hydroxy ether of bisphenol A); PHA, Poly(hydroxyalkanoate); PHB, Poly(3-hydroxybutyrate); PHBV, Poly(3-hydroxybutyrate-co-3-hydroxyvalerate); PLA, Poly(lactic acid); PP, Poly(propylene); PVA/PVOH, Poly(vinyl alcohol); PVC, Poly(vinyl chloride); RH, Relative humidity; S, Starch; SLNP, Solid lipid nanoparticle; TPS, Thermoplastic starch; WVTR, Water vapor transmission rate; WVP, Water vapor permeability; ZnO, Zinc Oxide; TiO₂, Titanium (II) oxide.

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recent years, it is anticipated that polymer packaging and coating in the food industry will continue to play an important role. Polymer films and coatings in food packaging applications can be divided into three main categories, as shown in Fig. 1. These are enhanced barrier, active, and intelligent/smart packaging materials, each with its unique functionalities. Barrier packaging's primary functionality is to provide moisture, gas, and aroma protection of packaged foods during production, transportation, storing, and selling.

On the contrary, active packaging materials provide functional attributes in addition to the barrier and are utilized to contain fresh and sensitive foods [3,4]. Furthermore, active packaging materials with antimicrobial and antioxidant properties can prevent the growth and contamination of microbes, such as fungi and bacteria, in food [5]. Thus, the freshness and quality of the food can be prolonged, further providing functional characteristics to enhance the protective properties of the packaging material. The third type of packaging material is intelligent/smart packaging. Such packaging materials are embedded with state-of-the-art sensor technology to monitor food conditions and display its information. These include smart features in the packaging that can measure the freshness of the foods and provide users with up-to-date information about products near-field communication (NFC) and quick response (QR) barcode scanning [3]. Unfortunately, the uses of these intelligent packaging films are still not widespread and only present in a few niche products.

Most of the commercial polymer packaging is derived from petrochemicals. The most common examples are poly(ethylene) (PE), poly(propylene) (PP), poly(amide) (PA), poly(vinyl chloride) (PVC), poly(ethylene terephthalate) (PET), and ethylene vinyl acetate (EVA) [6]. Although most of these synthetic polymer coatings and films possess many convenient and beneficial properties, they also impose major sustainability issues that jeopardize the environment and cause accumulating risks to consumers' health. The exceptional barrier properties of synthetic polymers such as gas, chemical, biological, or microbial resistance can be a double-edged sword as those resistance properties can make synthetic polymers extremely difficult to degrade at the final disposal step of the product life cycle [7]. Thus, most synthetic polymer-based packaging and coating materials end up in landfills, waste-to-energy facilities, or the general water and land environment after only a single-use [8,9]. It is essential to highlight that despite the ongoing COVID-19 pandemic, the

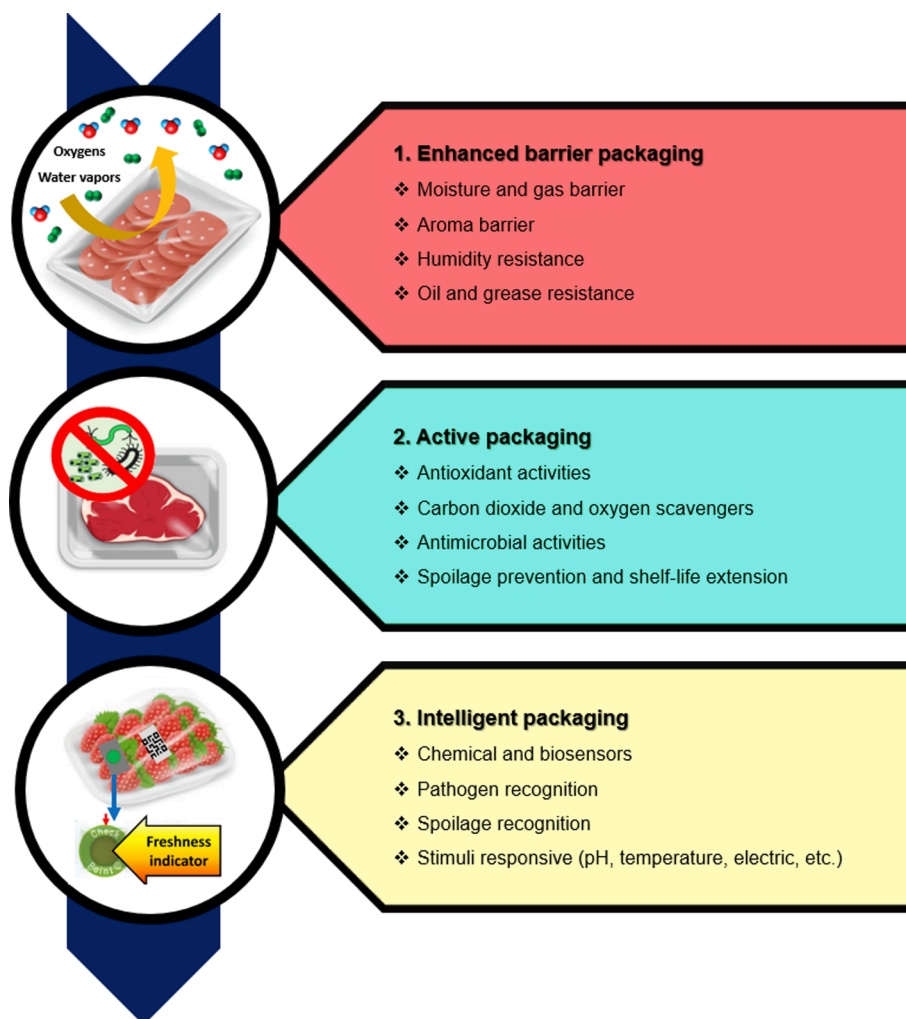


Fig. 1. Different types of functional packaging for food products.

production of traditional synthetic polymers or plastics in 2020 was estimated to maintain at the colossal number of 367 million, in which food service and packaging industries still constitute more than one-third of the total plastic usage [10,11].

For instance, a 2019 report from Deloitte Canada based on the data in 2016 indicated that only 9% of plastic wastes were recycled in Canada, while the remaining 86% ended up in landfills and 5% incinerated or littered in the environment [12]. Unfortunately, this is not so different in other jurisdictions. If the trend continues, the quantity of plastic waste accumulated in the environment could increase by up to one-third of the current waste by 2030. Moreover, some plastics can release harmful chemicals (e.g., Bisphenol A) and partially break down into microplastic, posing health and safety risks to humans and other animals. Thus, there is a desperate need to accelerate the design and development of sustainable alternatives to resolve plastic waste accumulation, the generation of microplastics, in conjunction with ecological challenges associated with the production of plastics [13–15].

Due to the increasing societal and environmental needs active and sustainable packaging material trends, natural macromolecules (or biopolymers), such as proteins, carbohydrates, polysaccharides, and lipids obtained from plant resources, agriculture and forestry residues, or marine sources have become popular for the fabrication of sustainable food packaging films. Unfortunately, such sustainable materials' success in packaging applications is curtailed by insufficient thermomechanical properties, high cost, and most notably inferior moisture and oxygen gas barrier properties [16]. Compared to traditional plastics, some biodegradable polymers (including polymers derived from natural resources or synthetic bioplastics designed to be biodegradable) demonstrate remarkable degradation in a short period into CO₂, H₂O, and benign residue in various environments via the action of microorganisms. Furthermore, unlike non-biodegradable synthetic polymers, which can sometimes release toxic residues causing health concerns for consumers, biopolymers generated from natural macromolecules are safe and can satisfy both health regulations and customer demands [17,18]. It should be noted that not all bio-based plastic or polymers are not necessarily biodegradable or sustainable, as many

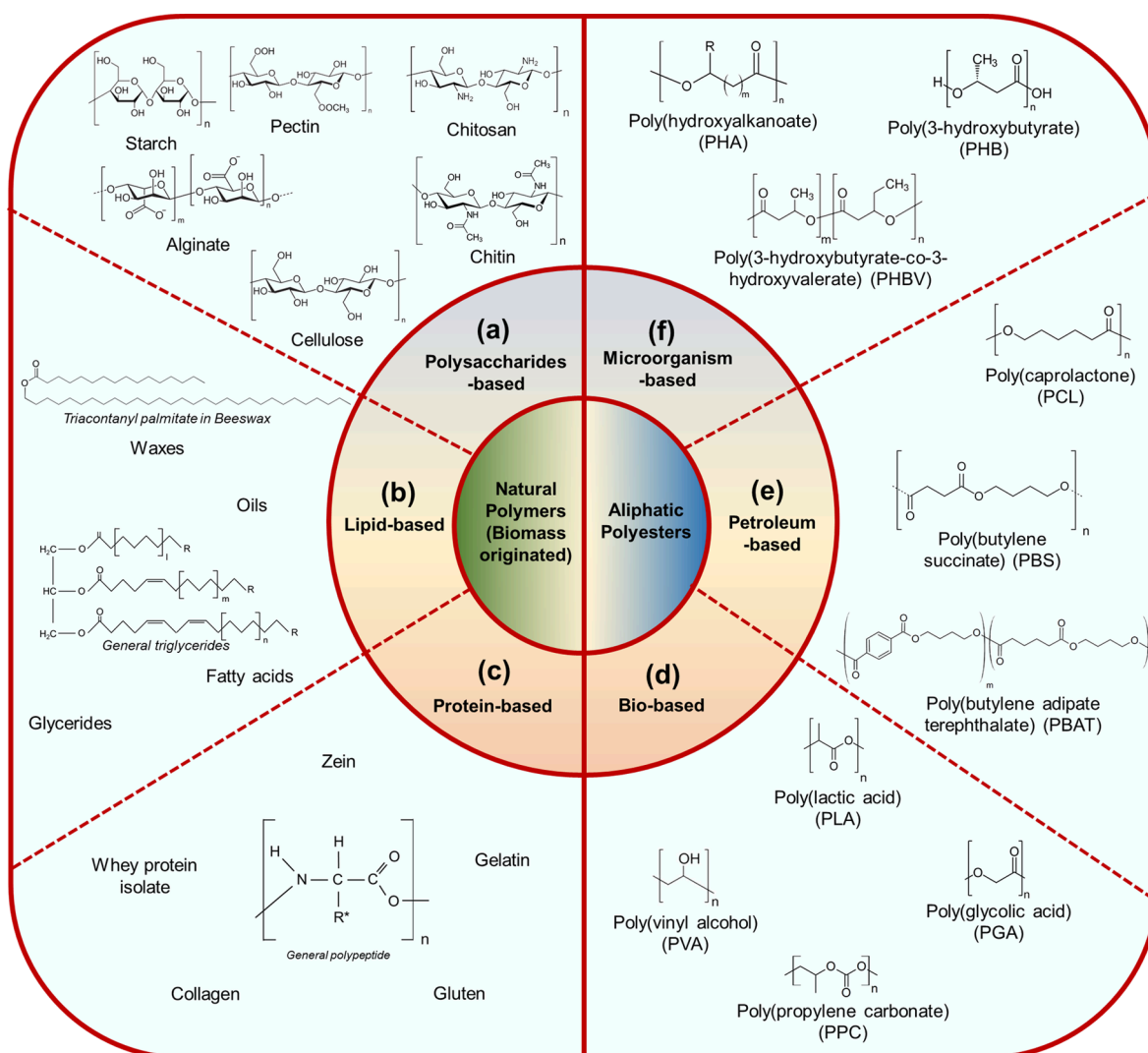


Fig. 2. Classification of biodegradable and/or compostable polymers employed for food packaging and coating applications.

other factors, such as chemical structures, synthesis processes (e.g., chemical crosslinking), additives, etc., can compromise their biodegradability, safety or sustainability [19]. Some examples of bio-based polymers that are not biodegradable includes bio-poly(ethylene terephthalate) (bio-PET), bio-poly(ethylene) (PE), or bio-poly(amide) (bio-PA) due to their carbon-packed structure preventing them from being biodegradable [20]. Thus, to highlight on the renewable aspect of high barrier multicomponent systems, this review will focus only on sustainable plastics that are either biodegradable, compostable or both.

Bioplastics used as a replacement for traditional plastics are classified into bio-based (e.g., poly(lactic acid) (PLA), poly(glycolic acid) (PGA)); petroleum-based polyesters (e.g., poly(butylene succinate) (PBS), poly(caprolactone) (PCL), poly(butylene adipate terephthalate) (PBAT)); and microorganism-derived (e.g., poly(hydroxyalkanoates) (PHAs), bacterial nanocellulose) (Fig. 2) [21–23]. Most of these bioplastics are compostable polyesters in various environmental conditions, which can also be easily processed with the existing polymer processing equipment like melt extrusion and injection molding, giving them advantages in terms of scalability. Besides these polyester bioplastics, the direct use of natural macromolecules with minimal modification demonstrated appealing effectiveness in food packaging and coating applications [24]. Nature derived macromolecules, such as polysaccharides (e.g., starch, cellulose, alginate, pectin, chitin, lignin, and chitosan) [25–27], lipids (waxes, oils, fatty acids, and glycerides) [28–30], proteins (e.g., soy, zein, canola, whey protein, gluten, keratin, and gelatin) [31–35] are some examples of highly biodegradable sustainable material options that attracted interest in packaging applications [33,36–38]. Nevertheless, unlike bio-polyesters, natural macromolecules are not easily processable, and scale-up operations are complex, limiting their potential usages in many food packaging applications. In general, bioplastics and biopolymers still have a few obstacles that need to be overcome, including high cost, unsatisfactory mechanical, thermal, and barrier properties to compete and displace the incumbent packaging plastics [39].

The strong drive for biodegradable polymers that provide good barrier properties against moisture and oxygen while simultaneously displaying good mechanical properties without compromising their biodegradability is ever-increasing, mainly for food packaging and coating applications [40,41]. Coatings and films need to have excellent barrier properties to promote appropriate isolation of food and minimize contact with oxygen (oxidation reaction, aerobic microorganisms' growth) or water (microbial growth, water activities, flavor change), as well as to avoid dehydration of moist foods, such as fresh meat, fruits, or vegetables [42,43]. Many strategies can be employed to improve the barrier properties of polymeric coatings. Most common approaches involve fabricating multiphase and multilayer film and coating materials through blending, surface coating, controlled assembling, layering, etc.

Thus far, there are only few comprehensive reviews on the moisture and oxygen barrier properties of packaging and coating materials. A recent review by Yu et al. reported the progress in the barrier, biodegradability, and recyclability requirement for natural chitin and cellulose-based packaging materials [44]. Another recent review by Wu et al., showed more general insights on the overall moisture/oxygen barrier properties of stand-alone films and summarized strategies to improve the permeation of biodegradable polymers, including chain architecture tailoring, crystallinity control, melt blending/multilayer co-extrusion, nanotechnology, and surface coating [45]. This review aims to present researchers with an in-depth understanding of moisture and oxygen gas barrier properties in renewable multiphase and multicomponent systems by providing the summary of the mathematical models of barrier properties and the factors that affect the permeability of polymer systems; as well as to report the most updated contributions on the improvement of the barrier performances of films and coatings strictly in the context of food packaging applications, such as blends, multilayer assembly, nanocomposites, emulsion food coating, and paper coating. Furthermore, challenges associated with renewable food packaging materials are also emphasized, as well as the potential of research that focuses on improving barrier properties is also discussed.

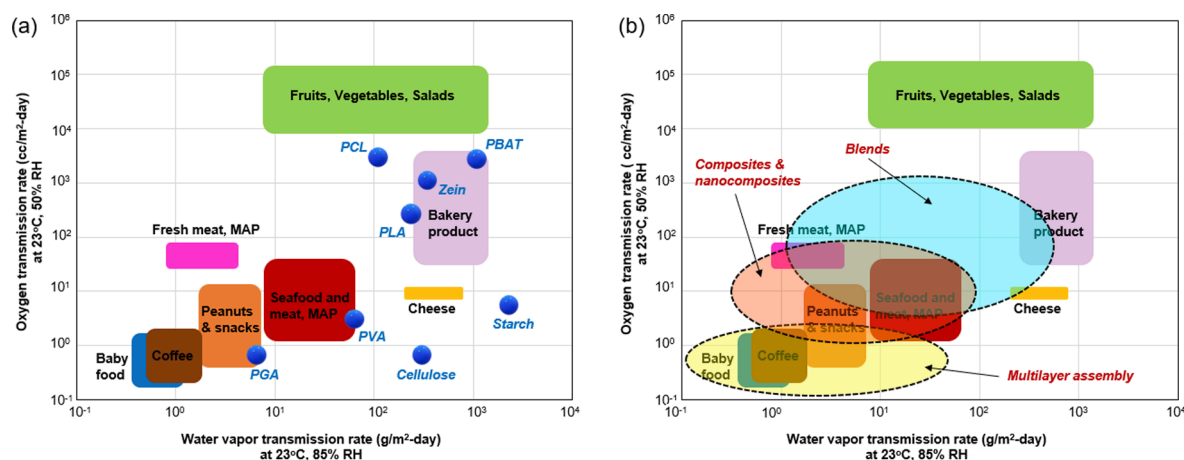


Fig. 3. Comparison of various food packaging water vapor/oxygen transmission rate requirement with (a) Individual polymers and (b) Multiphase polymer systems. Data obtained and reproduced from ref. [45,51].

2. Moisture and oxygen gas barrier properties

2.1. Barrier requirements of polymer coatings in various applications

Plastic packaging materials constitute a significant role in the economy by preserving food, pharma, biomedical goods [46,47]. The permeation and dissolution of moisture and oxygen gases through plastic packaging films (or layers) is directly correlated to the barrier capabilities of the constituent polymer. To be used as barrier films, wraps, or coatings, plastics must possess high barrier attributes to satisfy the strict protective standard in food packaging application. While the barrier properties of traditional polymers utilized in food packaging films and coatings are well established, such properties for the newer bioplastics are usually inferior and require further development and optimization. Comparison between packaging requirements and single-component biodegradable and bio-resourced materials are summarized in Fig. 3 (a & b). The boxes in Fig 3 indicate the normal moisture and oxygen permeability ranges for specific foods found in standard packaging.

In most cases, single stand-alone component biopolymer-based films and coatings could not satisfy the barrier requirement of packaging materials for several food products, such as instant coffee, cheese, or fresh meat. These polymers lack either moisture barrier or oxygen barrier properties and, in some cases, entirely lack both requirements. The only exception in the list of biodegradable polymers that provide both superior moisture and oxygen barrier properties is poly(glycolic acid) (PGA) [48]. Compared to the common polymers used in food packaging films and coatings (PET, PE, and PP), PGA exhibits superior moisture and oxygen gas barrier properties [48,49]. However, the production of PGA is much more complicated, energy-consuming, and costly [50]. As a result, PGA is not yet widely used in food packaging applications.

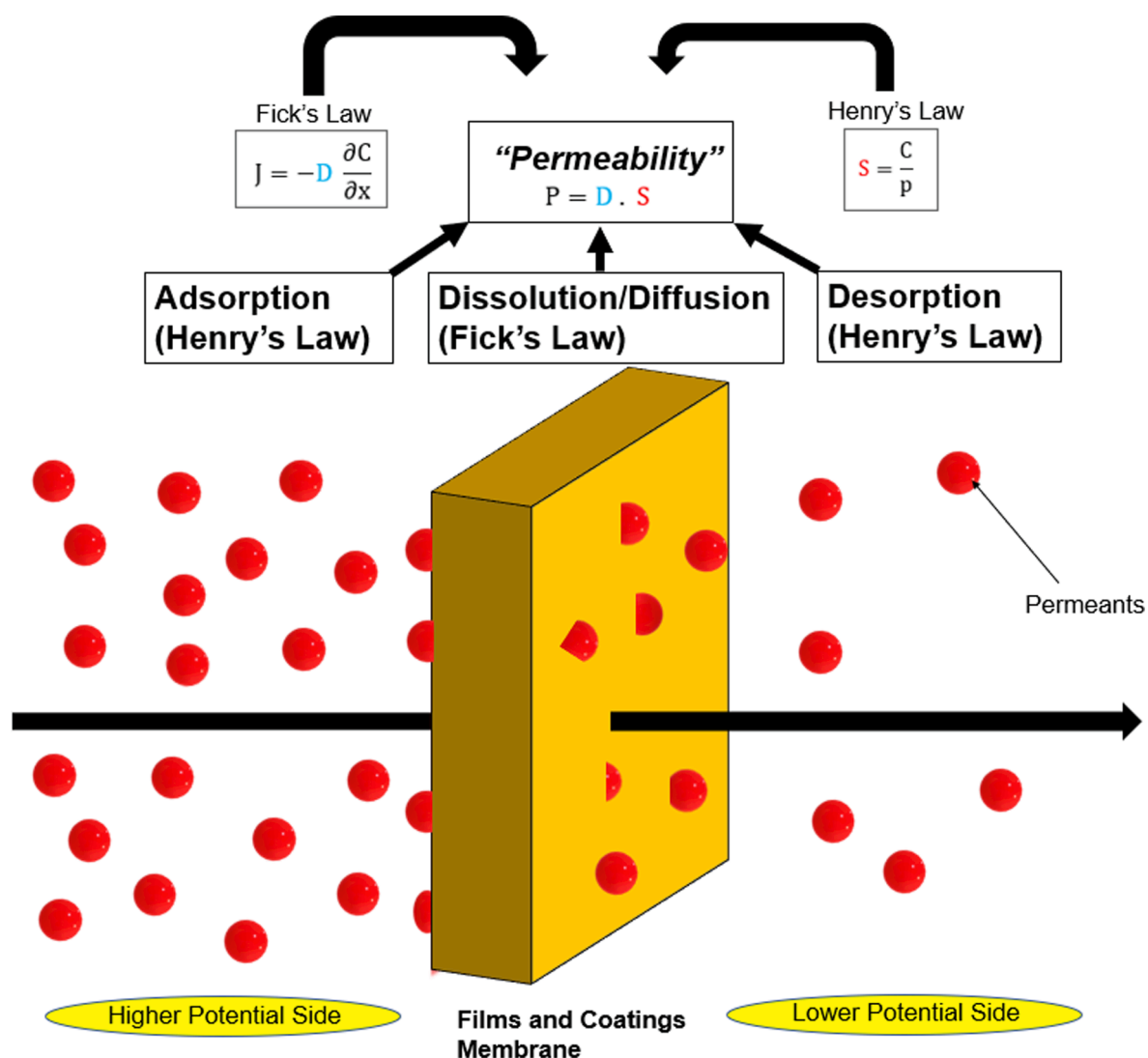


Fig. 4. Simple permeability model across a polymer film or membrane.

To obtain enhanced barrier performance, combining bioplastics with other traditional polymers, which possess excellent moisture and oxygen barrier properties, such as PET, PE, and PP, are common practices among many premier “green” packaging providers like Novamont (Italy), Biotec (Germany), and DuPont (USA) [52,53]. The combination of multiple materials in the form of blends, multilayers, and nanocomposites has shown to be effective strategies to meet industrial barrier standards, as the permeability ranges of these multicomponent systems satisfy most barrier requirements, as shown in Fig. 3b. However, incorporating non-biodegradable phase or component within the multiphase packaging coatings and films still constitutes a major challenge as the disintegration of such multiphase polymer systems with stark degradation differences leads to the generation of microplastics. Therefore, it is now widely accepted that such packaging films and coatings should be constructed from 100% biodegradable or compostable components to satisfy the sustainability requirement in addition to their physicochemical performance.

Promising developments are in progress on multicomponent and multiphase biodegradable systems that satisfy the barrier requirement of packaging materials, significantly reducing the overall moisture and oxygen transmission rate to the food products, as shown in Fig. 3b. For instance, recent works on TPS/PLA multilayer, TPS/PBAT multilayer, PLA/PPC blend [54], PBAT-hemp biocomposites [55], and starch ester/CNC nanocomposites [56] demonstrated promising barrier performance. However, barrier properties are highly dependent on the polymer's morphology, polarity, structure attributes, including the crystallinity, chain orientation, branching, free volume, and molecular interaction within polymer matrices [57]. To enhance the barrier properties of multiphase films and coatings, compatibilization techniques that involve chemical/physical modification, novel structure designs, incorporation of nanomaterials with specific morphologies are under intense investigation. This review briefly summarizes the key factors influencing the barrier properties of multiphase and multicomponent polymer films and coatings used in food packaging applications.

2.2. Models for permeations and barrier properties of polymer films and coatings

2.2.1. Simple transport mechanism

For polymeric materials, the permeation of moisture and oxygen gases involves the absorption capabilities of materials, diffusion, as well as desorption of these permeants from the films or coating surfaces, as shown in Fig. 4. The absorption of moisture or gas depends on the “solubility” of the permeants in the polymers. In other term, the ability to allow fluidic gas to pass through depends on the absorptive capabilities of the materials, which is dependent on factors such as degree of crystallinity (morphology), polarity, and molecular weight. In general, nature-derived biodegradable and renewable polymers have much worse barrier properties than those presented in petroleum-based polymers due to their inherent structures and chemical compositions. This is not surprising as these macromolecules are not purposefully designed for packaging applications as opposed to the synthetic counterparts used in packaging.

The most straightforward and most widely used mechanism of gas and moisture permeability is often illustrated and described as the idealized “Dissolution-Diffusion” (or “Solubilization-Diffusion”) model, which can be summarized into three main consecutive steps. The first step is adsorption, in which the moisture and oxygen gas molecules move on to one side of the membrane due to gradient potential difference of pressure, concentration or temperature. The second step is diffusion or dissolution. In this step, the molecules get solubilized inside the films or coatings and travel through the polymeric materials' void space (or free volume). The third and final stage is desorption, which entails escaping the diffused molecules on the other side of the film or coatings [58,59]. Fig. 4 shows a simplified mechanism of this permeation model. The movement of water vapor through a polymer layer or membrane can also be viewed as water molecules dissolving into the solid mass, travelling through the material, and finally evaporating on the other side of the layer or membrane. The diffusion/dissolution step primarily determines the absorbent properties of the material. Based on this mechanism, the permeability (P) can be related to the solubility (S) and diffusivity (D) coefficients as presented in Eq.1:

$$P = D \bullet S \quad (1)$$

The diffusivity coefficient term (D) can simply be described by either Fick's first (Eq.2) and second law (Eq.3), which describe the kinetic movement of low molecular penetrants (typically gas) across a polymer film or coating in terms of mass transfer proportional to the concentration gradient:

$$J = -D \bullet \frac{\partial C}{\partial x} \quad (2)$$

$$\frac{\partial J}{\partial x} = D \bullet \frac{\partial^2 C}{\partial x^2} \quad (3)$$

Where J is the flux of penetrant (amount of permeant passing through unit area per unit time), C is the concentration of the penetrant, and x is the one-dimensional direction of the concentration gradient. Fick's second law is used when the diffusion coefficient is independent of concentration, distance, or time. The diffusivity constant is dictated by both the polymer membrane and the diffusing permeant in terms of molecular size, temperature, pressure, and concentration gradients.

In order to integrate Fick's first law equation, a steady-state assumption must be implemented in the system. Additionally, the gas and vapor permeant concentration at equilibrium must also be kept constant on both sides of the film or membrane for simplicity. Thus, Eq. (2) can be written as:

$$\frac{Q}{A \cdot t} = J = D \bullet \frac{C_H - C_L}{L} \quad (4)$$

In this case, the flux J is expanded as the amount (Q) passes through a unit area (A) per unit time (t). C_H and C_L represent the

permeant concentration at higher and lower potentials. The denominator, L , denotes the thickness of the polymeric films or coatings.

The solubility coefficient term (S) can be determined from the ratio of the concentration of gas or vapor relative to the pressure in the gaseous state in the form of Henry's law:

$$S = \frac{C}{p} \quad (5)$$

Where C is the surface concentration of dissolved gas or vapor, and p is the partial pressure of the permeant gas. However, this method of determining the solubility model does not take the effect of plasticization of polymer film or membrane into account, which is not representative of polymeric materials in general. Therefore, in order to consider the plasticization effect, an extended dual sorption model to determine solubility adsorption of polymeric membrane matrix can be employed by the addition of additional terms

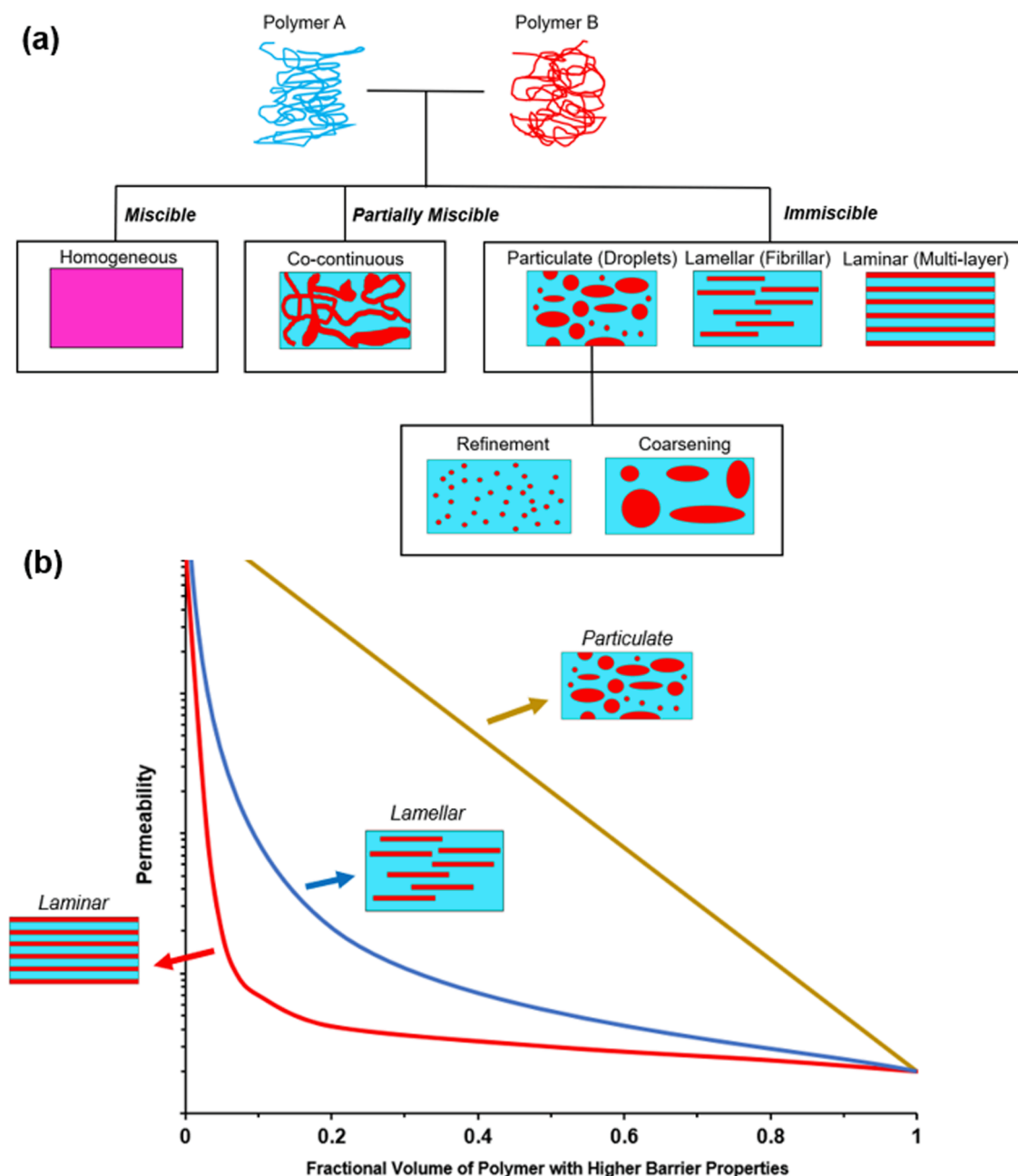


Fig. 5. (a) Different morphological categories of multiphase polymer blends. (b) Comparison of overall Permeability values between particulate, lamellar, and laminar system [68]. (Adapted with permission from ref. [44] from Wiley Online Library. Copyright ©2020.

for a concentration of dissolved gas or vapor (C), which is the combination of Henry's law and Langmuir's adsorption [60]:

$$C = C_H + \left(\frac{C_L b}{K_D} \right) \frac{C_D}{1 + bC_D/K_D} \quad (6)$$

where C_H and C_L are the concentration of dissolved gas or vapor defined by Henry and Langmuir solubility, b is the affinity of gas molecules being absorbed in the polymer, and K_D is Henry's law dissolution constant. The second term of this equation originates from the non-equilibrium nature of glassy polymers. The modified surface concentration will then be used to determine the solubility coefficient term through Henry's law (Eq.5).

Combining Eq.4 and Eq.5 and substituting them into Eq. (1), the permeability constant can be determined as [61]:

$$P = \frac{Q \bullet L}{A \bullet t \bullet \Delta p} \quad (7)$$

Δp represents pressure difference on the two sides of the polymeric films or coatings. This pressure difference is influenced mainly by gas type, humidity, and temperature. The dependence of permeability and transmission rate on temperature can be modified using the Arrhenius equation [62]. As seen in Eq.7, the permeability units for gases are generally expressed as volume-length/area-time-pressure. In the case of moisture permeability, the units are described as weight-length/area-time-pressure instead. Overall, these units of permeability and transmission rate can vary, depending on the measuring methods, e.g., "barrer" or "mol/m.s.Pa" for gas permeability and "perm" or "kg/m.s.Pa" for water permeability.

2.2.2. Developed models for multicomponent polymer films and coatings

In reality, the dissolution and diffusion of permeants in polymers are much more complicated to predict accurately. Especially in multicomponent systems, Fick's model is insufficient in predicting the solubility of multiple components as the theory neglects the interaction between permeants and polymer chains as well as one polymer chain from the other. Moreover, Fick's diffusion assumes a linear gradient concentration across the moving boundary of the films or coatings, which would only happen in an ideal situation [63]. However, the random orientations polymer matrix's free volume and swelling of polymer chains suggest that the concentration gradients behave more similarly to a non-linear model, attributed to the abruptness of permeants when they contact different polymeric segments multiphase components [64]. Therefore, it is essential to predict and develop dissolution and diffusion behaviors employing experimental modelling and characterization.

2.2.2.1. Multiphase blends and multilayer assembly. Blending is a common technique to produce barrier films and coatings that utilize limited amounts of the expensive high barrier polymer in an inferior and low-cost polymer matrix to fabricate a material with balanced performance and cost. The blends are often produced in a solvent-free and straightforward process, such as extrusion and injection process, making it a desirable technique in the food packaging industries. Since blending is the combination of two or more materials, the properties of polymer blend will often fall between those components. However, on rare occasions, the compatibilities and synergies between the blends can be so good that it might establish a system with improved properties that surpass its parent materials [65,66]. The barrier properties of any polymer blend depend strongly on the miscibility of the components. The morphologies of

Table 1

Mathematical models of moisture/ gas permeability through multiphase blends and multilayer coatings and films.

Blend system	Model	Equation	Eq. #
Particulate	Voigt [67,69]	$\ln P = \phi_d \ln P_d + \phi_c \ln P_c$ or $\ln P = \sum_{i=1}^n \phi_i \ln P_i$	(8)
Particulate	Maxwell [70]	$P = P_c \left[\frac{P_d + 2P_c - 2\phi_d(P_c - P_d)}{P_d + 2P_c + \phi_d(P_c - P_d)} \right]$	(9)
Particulate and co-continuous	Bruggeman [71]	$\left(\frac{P}{P_c} - \frac{P_d}{P_c} \right) \left(\frac{P}{P_c} \right)^{-1/3} = (1 - \phi_d) \left(1 - \frac{P_d}{P_c} \right)$	(10)
Particulate	Higuchi [72,73]	$P = P_c \left[1 + \frac{3\phi_d \left(\frac{P_d - P_c}{P_d + 2P_c} \right)}{1 - \phi_d \left(\frac{P_d - P_c}{P_d + 2P_c} \right) - K(1 - \phi_d) \left(\frac{P_d - P_c}{P_d + 2P_c} \right)^2} \right]; K=0.78$	(11)
Particulate and Lamellar	Lewis-Nielsen [74]	$P = P_c \left[\frac{1 + 2\phi_d \left(\frac{P_d - P_c}{P_d + 2P_c} \right)}{1 - \Psi \phi_d \left(\frac{P_d - P_c}{P_d + 2P_c} \right)} \right]; \Psi = 1 + \left[\frac{1 - \phi_m}{\phi_m^2} \right] \phi_d \phi_m = 0.785$ for square fibrillar packing, $\phi_m =$	(12)
Lamellar	Maxwell-Wagner-Sillar [75]	0.82 for random packing $P = P_c \left[\frac{nP_d + (1-n)P_c + (1-n)(P_d - P_c)\phi_d}{nP_d + (1-n)P_c - n(P_d - P_c)\phi_d} \right] 0 \leq n \leq 1/3$ for prolate ellipsoid, $n=1/3$ for spherical,	(13)
Co-continuous	Geometric Mean [76]	$1/3 \leq n \leq 1$ for oblate ellipsoid $P = P_c^{\phi_c} + P_d^{\phi_d}$ or $P = \sum_{i=1}^n P_i^{\phi_i}$	(14)
Laminar	Reuss [76]	$\frac{1}{P} = \frac{\phi_c}{P_c} + \frac{\phi_d}{P_d}$ or $\frac{1}{P} = \sum_{i=1}^n \frac{\phi_i}{P_i}$	(15)

Note: P_c and P_d are the permeability of the permeant in the continuous and dispersed polymer phases, respectively. ϕ_c and ϕ_d are the fractional volumes of the continuous and dispersed polymer phases, respectively

blending can be divided into three main categories: miscible, partially miscible, and immiscible (Fig. 5a). Miscible blend refers to mixtures of polymers with similar chemical structure and viscosity that can be combined into one homogeneous phase. It must be noted that achieving a miscible blend of two or more materials is a remarkably challenging task because of the unfavorable kinetic and entropy of mixing [67]. Thus far, there has been no report of permeation model development for biodegradable and renewable polymer blends with complete miscibility.

On the other hand, analytical models that predict the permeability of moisture and oxygen gases for binary immiscible blends have been extensively studied and well-presented over the last few decades, especially on the particulate (dispersed droplets) and lamellar (fibrillar) distribution of a polymer in another polymer matrix. A number of reviews, such as Aroon et al. [77], Gonzo et al. [78], and Wu et al. [79], have investigated and compiled various possible mathematical models that would be fitting to describe polymer blend systems. Among the developed models presented in these reviews, a few important and common mathematical equations are selected and summarized in Table 1.

Various approaches are developed to take the permeability for the particulate distribution of dispersed phases in polymer films or coatings into account. The maximum permeability of immiscible blend coating can be described by an equation analogous to Voight model from the Rule of Mixture (Eq. 8) [69]. This predictive model is used when the dispersed phases are parallel to the direction of permeation flux, which gives the Voight model another name, known as the Parallel model. Since the Voight model acts as the upper boundary for permeability, all the developed models should predict lower permeability values. Among the equations, the most widely accepted model to depict the permeability in polymer blends is the equation proposed by Maxwell. Maxwell's model was initially used to estimate the dielectric conductivity properties of polymer blends, but the formula was later developed into an equation to predict permeability, as shown in Eq. 9 [70]. The limitation of this model is that it is adequate only in the prediction of spherical refinement morphology of the blended polymer matrices and only with dispersed volume fraction (ϕ) < 0.2. Many other analytical models were adapted from the Maxwell equation and were extensively investigated, with the most prevalent examples being Bruggeman, Higuchi, Lewis-Nielsen, and Maxwell-Wagner-Sillars. Bruggeman expressed the permeability of a particulate system blend by using an effective medium theory approach in (Eq. 10), which can be applied for both particulate and co-continuous systems [71]. This equation is appropriate when there is a small difference in the permeability of two polymer phases, as well as for compositions near 50%.

The formula expressed by Higuchi (Eq. 11) was derived from experimental data and is applicable to predict particulate blend systems regardless of distributions or sizes, including refinement and coarsening morphology [72]. From dielectric constants collection, Higuchi determines the empirical constant (K) for spherical systems to be 0.78. Lewis and Nielsen experimentally developed their own model (Eq. 12) by exploring spherical particulate model's relative elastic modulus and volume fraction [74]. This model considers the maximum packing fraction (ϕ_m) and can predict a square lamellar distribution system's permeabilities by setting ϕ_m to 0.785 [80]. For the specific case of dilute dispersion of ellipsoid fillers in lamellar orientation, the modified Maxwell-Wagner-Sillars equation (Eq. 13) can be applied with a shape factor (n) $\leq 1/3$ [75]. The expression will be reduced to the Maxwell equation when $n = 1/3$. Based on these models proposed for the lamellar system, the P values of such morphologies would typically fall between particulate and laminar system values, as observed in Fig. 5b.

Laminar blends refer to a system of complete immiscible morphology between polymer blends that are separated into different layers in the perpendicular direction to the route of molecular diffusion. This laminar system is generally regarded as a multilayer

Table 2

Mathematical models of moisture/ gas permeability through nanoplatelets-type nanocomposites. (Adapted from Choudalakis and Gotsis [83], Elsevier copyright ©2009).

Array/Orientation (cross-section)	Model	Equation	Eq #
All	Percolation threshold [78]	$P = P_c(\phi_d - \phi_c)^t$	(17)
Intercalated	Nielsen [85]	$P = P_c \left(\frac{1 - \phi_d}{\tau} \right); \tau = 1 + \frac{\alpha}{2} \phi_d; \alpha = \frac{L}{W}$	(18)
Intercalated	Cussler [86,87]	$P = P_c \left(1 + \frac{\alpha^2 \phi_d^2}{1 - \phi_d} \right)^{-1}$	(19)
Intercalated	Moggridge [88]	$P = P_c \left(1 + \frac{2}{27} \frac{\alpha^2 \phi_d^2}{1 - \phi_d} \right)^{-1}$	(20)
Exfoliated	Bharadwaj [89]	$P = P_c \left[\frac{1 - \phi_d}{\tau} \right]; \tau = 1 + \frac{\alpha}{3} \left(\mu + \frac{1}{2} \right) \phi_d; \mu = \frac{1}{2} (3 \cos^2 \theta - 1)$ $\mu = -1/2$ for parallel orientation, $\mu = 0$ for random direction, $\mu = 1$ for perpendicular orientation	(21)
Aggregated (Phase separated)	Nazarenko [90]	$P = P_c \left(\frac{1 - \phi_d}{1 + \frac{\alpha}{2N} \phi_d} \right)$	(22)
Exfoliated and Aggregated (Phase separated)	Combined Bharadwaj-Nazarenko [83]	$P = P_c \left[\frac{1 - \phi_d}{1 + \frac{\alpha}{3N} \left(\mu + \frac{1}{2} \right) \phi_d} \right]$	(23)

Note: P_c and P_d are the permeability of the permeant in the continuous and dispersed polymer phases, respectively. ϕ_c and ϕ_d are the fractional volumes of the continuous and dispersed polymer phases, respectively. ϕ_t is the fractional volume at percolation threshold and t is the critical exponent. τ is the tortuosity and α is the aspect ratio of nanoplatelets. μ is the orientation factor; and N is the number of layers in a layer stack.

assembly.

A partially miscible blend system typically refers to the morphology of the co-continuous distribution. Since a network system of co-continuous system is randomized, it is especially challenging to develop a specific system that can accurately predict the moisture and oxygen gas permeability values based on the orientation of the dispersed matrices. Predicting the permeability of co-continuous (or lamellar) blend can be estimated by utilizing a simple Geometric Mean model, presented in Eq. 14 [76]. This Geometric Mean model assumes random distribution of phases and takes the weighted geometric mean to calculate polymer permeability.

The permeability of multilayer films or coating (laminar) systems can be calculated through the lower bound of the permeability model described by the Reuss model from Rule of Mixtures (Eq. 15) [76]. Comparison of the permeability of the ideal multilayer system to those of the blend system is illustrated in Fig. 5b. It is evident that multilayer assembly would achieve enhanced barrier properties with less amount of high barrier polymers, indicating it is a superior approach to improving the barrier properties of renewable and sustainable materials.

2.2.2.2. Fillers and nanocomposites. With the current interest in nanotechnology, nanocomposite materials have attracted substantial research interest for packaging films and coatings. Different modelling approaches have proved that barrier properties can significant improvement improve with high aspect ratio nanoparticles [81] in polymer matrices. The leading example is the Nielsen model (Eq. 18) as shown in Table 2. Over the years, nanoparticles' permeability models have been proposed for three main categories of nanofillers, depending on their shape. Iso-dimensional particles, which have unity dimension in all lattice directions (e.g., silica, spherical CNC, TiO_2); elongated particles, which consist of fibrils with diameter < 100 nm but with much higher length up to hundreds or thousands nanometers (e.g., platelets CNT, CNF, collagen); and nanoplatelets (layered) particles, which consist of flat-shaped particles with thickness ranging from 1-10 nm (e.g., graphite, nano-clay, CNC) [82]. Categories of nanoparticles are shown in Fig. 6a.

The models presented in Table 1 for blends can also be applied for the permeability's of isometric nanoparticle filled composites,

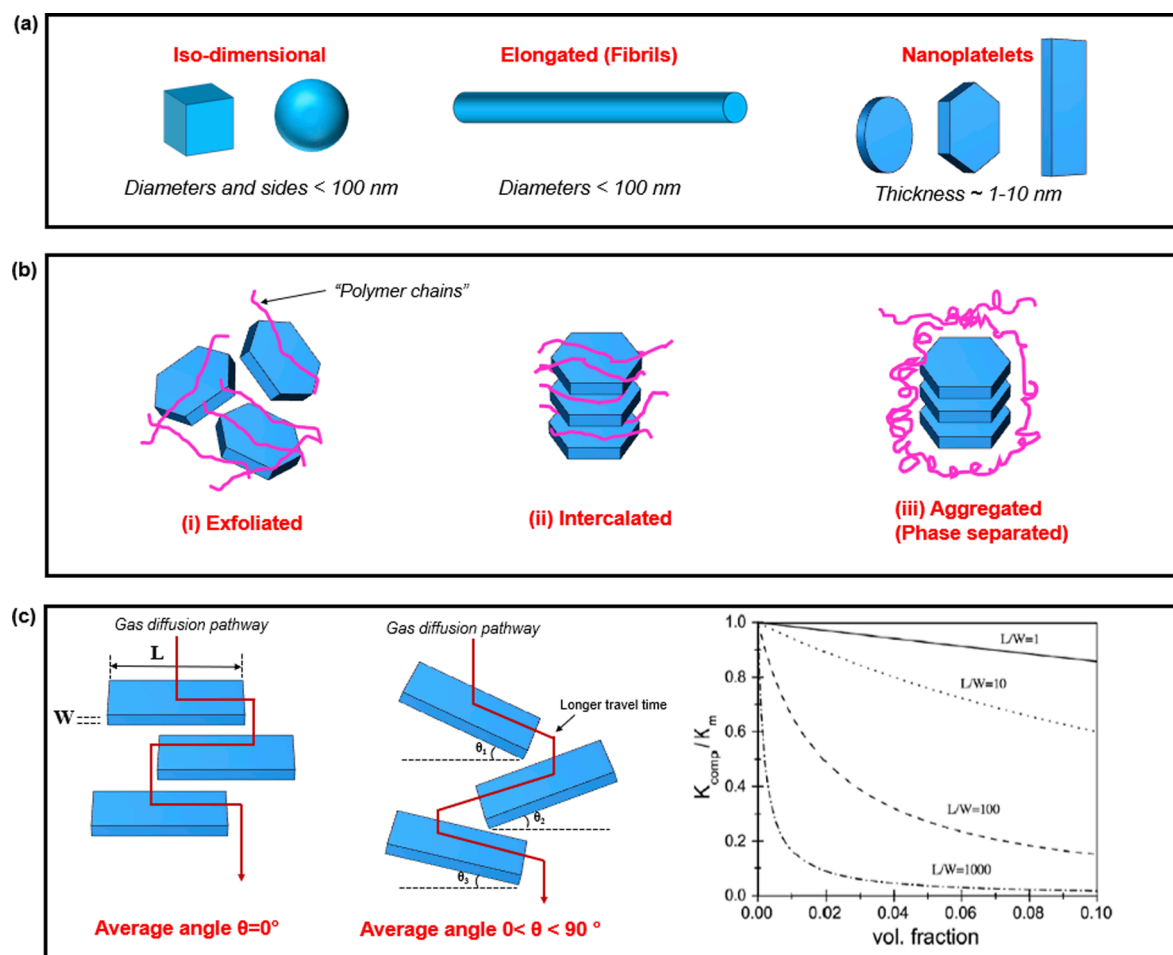


Fig. 6. (a) Different shapes of bio-based nanoparticles. (b) Formation of exfoliation, intercalation, and aggregation of nanoplatelets in polymer matrices. (c) Schematic on the orientation influence of tortuosity on the network of layered nanoplatelets. Influence of aspect ratio on the relative permeability of nanocomposites (as shown by Nielsen's model); K_{comp}/K_m stands for relative permeability [83] (Copyright ©2009. Reproduced with permission from Elsevier)

namely **Eq. (9-12)** for spherical shape and **Eq. (10 & 12)** for cuboid shape nanoparticles. However, in the nanoparticle network dispersed uniformly in polymer matrices, the fillers would be considered impermeable due to nanoparticles' high crystalline structures and high aspect ratio. Thus, the aforementioned models would assume that the permeability of the disperse phase (P_c) = 0.

In the case of nanotubes in polymer matrices, the second generalized Maxwell equation can be used:

$$P = P_c \left[1 + \frac{(1 + G) - \phi_d}{\left(\frac{P_d/P_c + G}{P_d/P_c - 1} \right) - \phi_d} \right] \quad (16)$$

where G is the geometric factor accounting for the shape of the dispersants, which can be determined experimentally. This model was adapted by Ge et al. [84] and used for the gas permeability through the CNT-polymer matrix membrane structure.

Among all three types of nanoparticles, nanoplatelets have been shown to considerably improve the barrier properties inside a polymer matrix. Thus, significantly more models are developed to predict the permeability of this type of nanoparticles than those of iso-dimensional and elongated particles. A review by Choudalaski and Gotsis provided an in-depth description of the models for the permeation of nanoplatelet systems in polymer. The formula can be generalized to other nanoplatelet systems [83]. For platelet-shaped nanoparticles, there are three main types of morphologies: exfoliated, intercalated, and aggregated (phase-separated), which can be observed in **Fig. 6b**. The equations used to describe the moisture and oxygen gas permeation through these three nanoplatelet morphologies are outlined in **Table 2**. These models typically assume the nanoparticles have either parallel orientations perpendicular or random orientation at average angles (θ) to the direction of diffusion, as illustrated in **Fig. 6c**.

The first permeability model for nanocomposites was developed based on percolation theory. Thus, a simple power-law can describe the relationship between the nanocomposite permeability within the critical percolation threshold, as shown in **Eq. 17** [78]. Another simple permeability model for the parallel arrangement of rectangular platelets with thickness, W and length, L are presented in **Eq. 18** [85]. This expression is the Nielsen model, and it is one of the most basic equations to predict the permeability of polymer matrix with nanoplatelets acting as impermeable barriers to the diffusing permeant molecules. This formula accounts for both the aspect ratio of nanoparticles and the tortuosity provided by the arrangement of platelets. It should be noted that the tortuosity is correlated to the aspect ratio of nanoparticles; thus, a change in aspect ratio would directly affect the permeability through the nanocomposites. The significant effects of aspect ratio and tortuosity on the diffusion of permeants are illustrated in **Fig. 6c**. The overall diffusion of gases through random arrays of single laminae was intensively studied by Cussler's group [88,91,92]. They extended Nielsen's model and proposed a different formula (**Eq. 19**) that connects the diffusion through the film or coating with the pore/slit distance between particles [86]. Compared to Nielsen's model, Cussler's model showed a substantial reduction in permeability at smaller nanofiller volumetric fractions due to the more restrictions it accounted [86]. In order to consider the permeability of hexagonal flake nanoparticles arranged in parallel arrays (i.e. nanostarch, silica, or ZnO), Moggridge et al. gave an extension of the Cussler model and added a specific hexagonal platelet-shape coefficient (2/27), as indicated in (**Eq. 20**) [88]. This coefficient suggested that hexagonal shape nanoparticles objectively have barrier properties reducing effect in the nanocomposite systems compared to regular rectangular shape ones.

The assumption of parallel array orientations of nanoplatelets can lead to underestimating actual moisture and oxygen gas permeability values. To account for different geometry and orientation of the dispersed nanofiller phase inside matrix coatings, Bharadwaj proposed a two-dimensional model that was another modified version of Nielsen's model (**Eq. 21**) [89]. This model provided more considerations for the physical orientations of the nanoplatelets fillers (μ), which can be beneficial in predicting the permeability performance of exfoliated polymer structures. It is highly suggested that the orientation at angles will force permeants to take longer tortuous pathways to travel across the film/coating (**Fig. 6c**); thus, reducing the overall moisture/oxygen barrier properties. Based on the developed formula, Bharadwaj concluded that the dispersion of nanoplatelet fillers at a perpendicular orientation to permeation direction would give the highest barrier performances, indicating that intercalated structures are preferable exfoliated structures in terms of obtaining enhanced barrier properties.

In many cases, the permeability of renewable and biodegradable nanocomposites can be affected by the aggregation of nanoplatelet particles due to poor compatibility between the fillers and polymer matrices. The aggregation can especially happen at high concentrations of nanoparticles and can act as a low resistance pathway for moistures and gases transport within the composites. A model proposed by Nazarenko et al. (**Eq. 22**), derived from Nielsen's permeability equation, considers the aggregated morphologies present in the matrix by providing the term N to represent the number of layers in the layer stack [90]. The larger the value of N is, the larger the degree of aggregation, resulting in deficient barrier performances. One limitation in Nazarenko's formula is that it assumes the stacked nanoplatelets all have a perpendicular orientation to the flux direction. To account for the random orientation of the aggregated layers, an expression (**Eq. 23**) was developed from a combination of (**Eq. 21 and Eq. 22**) [83]. This expression is versatile, as it can calculate permeability as a function of volume fraction, geometry, orientation, and degree of orientation.

3. Factors affecting barrier properties of multicomponent renewable polymer coatings and films

Most renewable and sustainable materials are susceptible to the permeation of moisture and/or gases, however, to a very different extent, primarily dictated by the intrinsic nature of either the polymers or the permeants. Thus, to fabricate high barrier packaging coatings and films, many research focuses on figuring out how the attributes of the polymer materials can affect the barrier properties. It has been widely reported that the barrier performance of renewable food packaging coatings and films can be influenced by multiple factors, such as chemical structures, polymer architecture and morphologies, polymer crystallinity and orientation, as well as the

combination of different systems like polymer blend/multilayer assembly or the incorporation of nanofillers and other additives. Understanding the fundamental basis and the prominent impacts of these factors on the permeability performance of biodegradable packaging materials is crucial for future development and practical purposes to high barrier sustainable packaging materials. Parameters such as polarity, free volume, intermolecular cohesion, and cross-linking are important factors that determine the barrier properties of the particular polymer systems. These factors are highly dependent on each other, and it is important to consider their interactions when selecting appropriate materials for high barrier packaging coatings and films.

3.1. Chemical structure and polarity

One of the most important factors that considerably contribute to the barrier properties of sustainable polymers is the inherent chemical structures and the resulting polarity of the polymer chains. As mentioned in the mathematical models, a significant aspect that contributes to the permeability through polymer coatings or films is the ability of the material to absorb penetrating gases or vapor [93]. The “solubility” of the permeant is highly dependent on the chemical structure and polarity of the polymer and the constituting monomers [59].

In polymer science, the study of surface functionality/polarity and chemical interaction is extensively studied to understand the overall functional performances of non-biodegradable and biodegradable polymer systems, including but not limited to the mechanical, physical, and thermal as well as moisture and oxygen gas barrier properties. The chemical interaction between the permeant fluids and polymeric chains can affect packaging materials’ permeability and barrier performance, which is especially more prevalent for moisture vapor permeability. It is widely accepted that the water permeation activities of hydrophilic polymers and fillers are significantly higher than hydrophobic polymers and fillers [94–96]. Specifically, bio-based polymeric materials tend to contain much more polar functional groups than traditional petroleum-based polymers that subsequently make these polymers much more susceptible to moisture attack via polar-polar interaction and hydrogen bonding. Thus, surface tailoring and modification are typically implemented as a stratagem to either (i) improve and optimize the properties of existing polymers, (ii) enhance the interaction between the compounds in multicomponent system, or (iii) introduce desirable functional attributes to the polymer. Representative examples are polysaccharide-based materials, such as starch or cellulose are the prime subjects of surface modification to improve the mechanical and barrier properties in composites or blends [97,98]. This is attributed to the abundant hydrophilic hydroxyl (-OH) group present on the polymer backbone, allowing easy moisture association via hydrogen bonding, and thus, require further

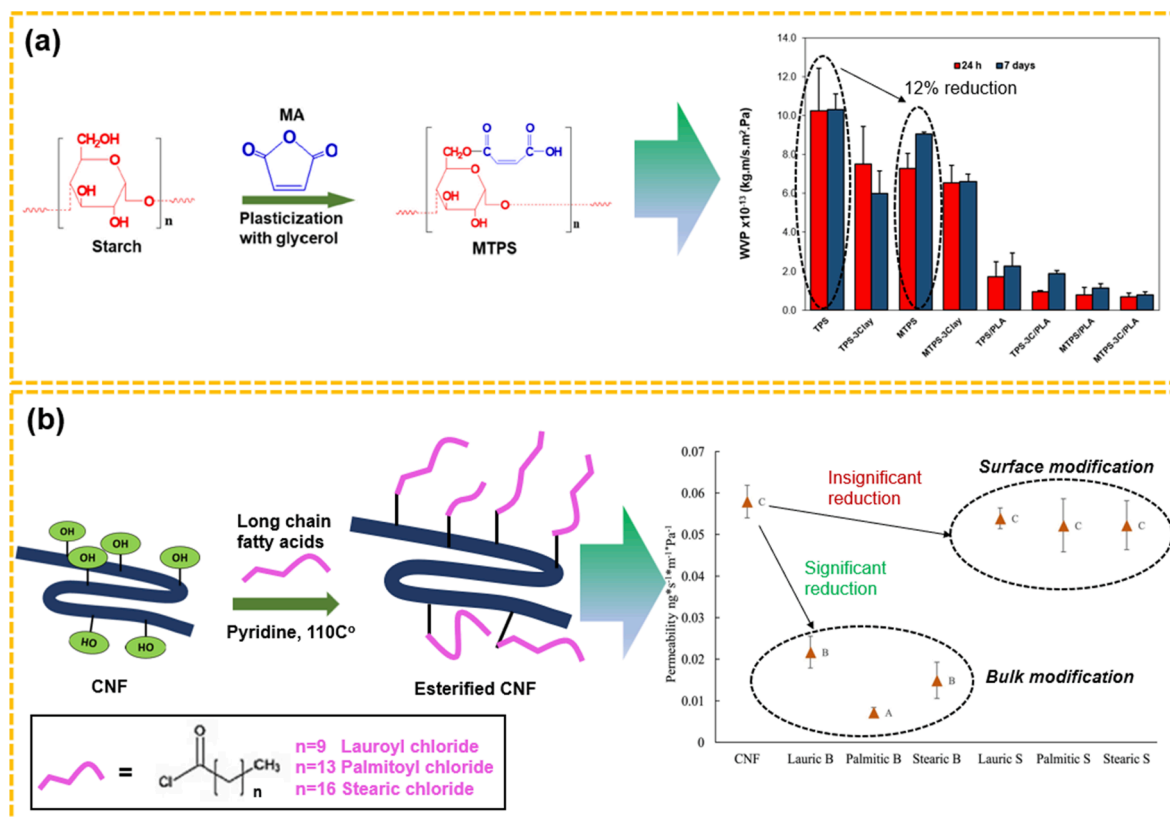


Fig. 7. (a) Maleation reaction of TPS films and their effects on WVP. Adapted with permission from ref. [99] from Elsevier. Copyright ©2021. (b) Esterification reaction of CNF films with different fatty acids and their effects on WVP. Adapted with permission from ref. [105] from Elsevier. Copyright ©2020.

modification [16]. On the other hand, these hydrophilic polysaccharide-based materials provide an excellent barrier against nonpolar O_2 and other similar non-polar gases [99–101].

A multitude of researches have been conducted to partially substitute the $-OH$ functional groups of cellulose with more hydrophobic functional groups to improve the moisture barrier properties, with various degrees of success [99,102,103]. Most hydrophobic chemical modification of packaging coatings and films involves tailoring and controlling the polymer structure on the surface, which does not change the fundamental backbone and can keep most of their physicochemical and biochemical properties. For example, Li et al. synthesized hydrophobic acylated cellulose nanofibrils (CNF) films using 10-undecylenoyl chloride modifier followed by solvent casting [104]. The authors observed that the water contact angle (WCA) has substantially increased (from 48.2° to 88.1°), which corresponded to a WVP reduction of about 62.4%, indicating a significant improvement in terms of water vapor barrier properties of the modified CNF based films. Another study carried out by Trinh et al. observed that the surface grafting of maleic anhydride slightly reduced TPS's water vapor permeability (WVP) values by about 12% (Fig.7a) [99]. However, the surface modification did not intend to minimize the WVP of TPS directly but rather to improve the barrier properties by promoting the TPS's interaction with hydrophobic PLA in a multilayer assembly. Similarly, an investigation by Balasubramaniam et al. showed that the bulk modification of CNF with different fatty acids (lauroyl, palmitoyl, and stearoyl chloride) caused a massive reduction in the WVP of the resulting films, while the surface modification of the CNF with all the three fatty acid reagents did not have a significant effect on the WVP of the biopolymer films (Fig.7b) [105]. Due to the limitation in the reaction sites of surface grafting, it should be expected that barrier properties can only be marginally enhanced, though not as much of an improvement as other methods such as blending, multilayer construction, or composites. On another note, it must also be acknowledged that changing the chemical structures of biodegradable polymer backbones or monomers could potentially eliminate the desirable biodegradability traits of renewable coatings and films [45]. As suggested by Witt et al., the biodegradability of materials depend on the chemical structure rather than the feedstock [106].

In terms of natural hydrophobic biopolymers, some proteins, such as zein or keratin, are often employed to improve the WVP of packaging films due to their hydrophobic structures provided by the non-polar amino acid residues or their structural packing [107,108]. Zein, extracted from corn, are known to be insoluble in water and contain a large amount of hydrophobic amino acid domains (leucine, proline, and alanine) [109,110]. Fabra et al. developed a multilayer structure which include high barrier interlayers of zein nanofibers on PHBV films to enhance the water vapor barrier of the resulting films [111]. This improvement of WVP was attributed to the inherently hydrophobic properties of zein as the water vapor barrier enhancement become more apparent at higher concentrations of zein were electrospun as nanolayers. Similarly, keratin, found in mammals' hairs or bird's feathers, is also hydrophobic and insoluble in water because of the packaged molecular structure as well as the prominent hydrophobic cysteine (sulfur-containing) amino acid composition [38,112]. Thus, keratin has been crosslinked or incorporated in other biopolymers as supporting materials to improve water vapor barrier of multicomponent systems. For instance, keratin was combined with dialdehyde carboxymethyl cellulose (DCMC) and PVA to generate DCMC/keratin and [108] PVA/Keratin blends [113], respectively.

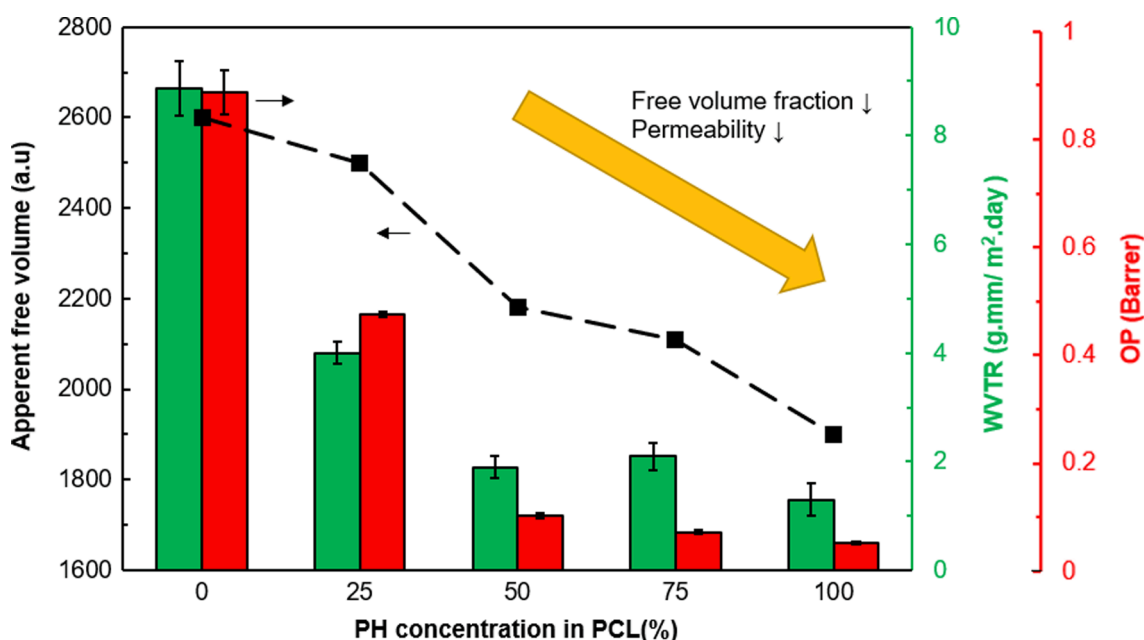


Fig. 8. Reduction in WVTR and OP of PCL/PH blends as free volume decreased with the higher loading of PH in the matrices [116] Adapted with permission from ref.[116] from Elsevier. Copyright ©2020.

3.2. Polymer architecture and morphological factors

The polymer architecture and morphology factors also play a crucial role in controlling the overall barrier properties of renewable polymers. The permeability of polymer coatings and films can also be affected by the “diffusivity” element. Another physical and morphology factor that affects the diffusivity properties of renewable coatings and films is the free volumes and intermolecular cohesion. Free volume in any polymeric material is defined as the internal space available inside the polymer that would allow polymer chains mobility. This available space within polymer matrices can act as open “pathways”, which ultimately allows diffusive activities of penetrants across polymeric films and coatings [114]. Free volume in a polymer can stem from the addition of plasticizers, which typically are incorporated in biopolymers (e.g., glycerol and other polyols in thermoplastic starch and protein) to increase the free volume among polymer chains and ease processability, which subsequently improves the elasticity and flexibility of the final packaging coatings and films. However, plasticizers reduce the glass transition temperature (T_g) and worsen permeability as a tradeoff.

It is also noted that the intermolecular cohesion of polymer chains directly affects the barrier performance of polymer matrices. Typically, high intermolecular cohesion energy between polymer chains improves the barrier transport against moisture and oxygen gases penetrants. This is because higher intermolecular cohesion energy decreases the free volume within polymer by restricting polymer chain mobility and consequently reduces the voids and diffusive mobility of permeant gases or vapor [115]. A recent study by Corres et al. elucidated the role of free volume in the barrier properties of films using PCL with Poly(hydroxy ether of Bisphenol A) (PH) models. The study provided good insight on the topic, which can be extended to other renewable polymer systems [116]. In the study, both the water vapor transmission rate (WVTR) and oxygen permeability (OP) values of PCL significantly reduced; in other words, enhanced the barrier properties with the increase in the PH concentration. This was attributed to the reduction in the free volume sizes contributed by PH (Fig. 8). Though the free volume size and barrier properties seem to be directly correlated, the authors also suggest that the chemical compatibility between PCL and PH plays vital part in reducing the overall permeability.

The barrier performance of polymers can also be improved via chemical crosslinking that involves the covalent linking of two or more molecular chains using a crosslinking agent. Crosslinking is an appealing technique that can limit the mobility of polymer chains and decrease the free volume within the polymer structures, which can lead to the reduction in oxygen uptake and water activities of biomass-derived polymers, such as polysaccharides or proteins [117]. However, crosslinking is especially favorable for protein-based plastics, such as gelatin, because protein-based materials have abundant primary functional groups, such as amines, carboxyl, hydroxyl, and sulfhydryl, that are amenable to form covalent. Most studies showed that a higher crosslinking degree can improve moisture and oxygen resistant with structural flexibility tradeoff [16,118]. Muñoz et al. mentioned that the use of aldehyde crosslinking agents such as formaldehyde, glutaraldehyde, and glyoxal all reduce the WVP values by up to 30% after crosslinking

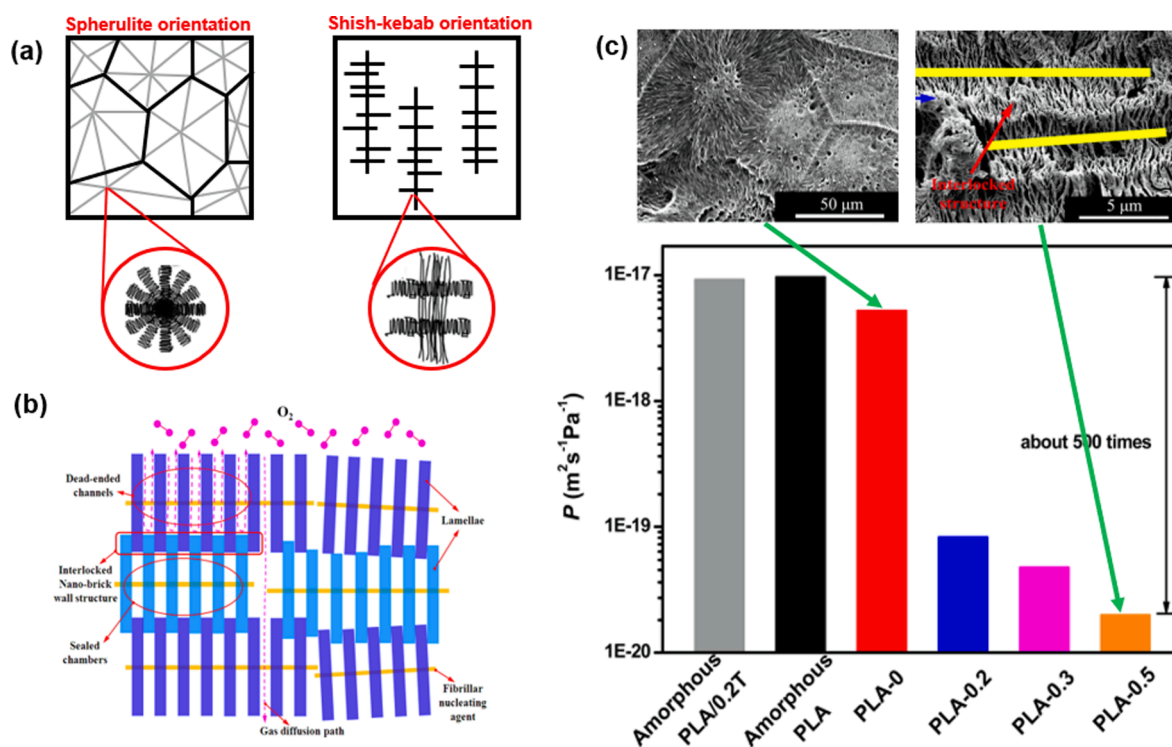


Fig. 9. (a) Different crystalline structure in semicrystalline polymers. (b) Mechanism of the interlocked parallel “shish kebab” structure that inhibit oxygen permeation and (c) The improved oxygen barrier of PLA with the parallel “shish-kebab” structure. PLA-(0.2-0.5) indicated samples including nucleating agent N,N,N' -Tricyclohexyl-1,3,5-benzene-tricarboxylamide (TMC-328) from 0.2 to 0.5 wt.% in PLA resins processing, respectively. [124] Reproduced with permission from American Chemical Society. Copyright ©2014.

treatments. However, the WVP is still poor compared to synthetic plastics used in packaging applications [119]. A recent study by Gao et al. showed that the crosslinking of zein protein films with glutaraldehyde could produce packaging materials with superior oxygen barrier properties that can even compete with commercial PE and PP [120]. However, crosslinking does not always improve the barrier properties of biodegradable polymer coatings or films. Depending on the crosslinking agent, crosslinking can have a detrimental effect on the materials' moisture and oxygen gas barrier properties in some cases. Uranga and Leceta reported an increase in the WVP values of their fish gelatin crosslinking with citric acid, attributed to the unreacted hydrophilic citric acid $-\text{COOH}$ remaining inside the structure of the films and absorbing more moisture [121]. The correlations between crosslinking and moisture and oxygen barrier properties are often unpredictable in practice, and thus, investigations through experimental designs are required to confirm such relationships.

3.3. Polymer crystallinity and molecular orientation

For the last few decades, polymer crystallinity and molecular orientation have been observed to play a major role in controlling the barrier properties of renewable polymers. Overall, the crystalline phase of polymers is considered impermeable due to the chain immobilization effect, and the amorphous phase primarily dictates the barrier properties. In addition, the lamellar crystals of semi-crystalline polymers reduce the amorphous phase chain mobility that further reduces the diffusion rate of permeants. Previous investigations on synthetic plastics have confirmed the validity of the relationship between the degree of crystallinity and barrier

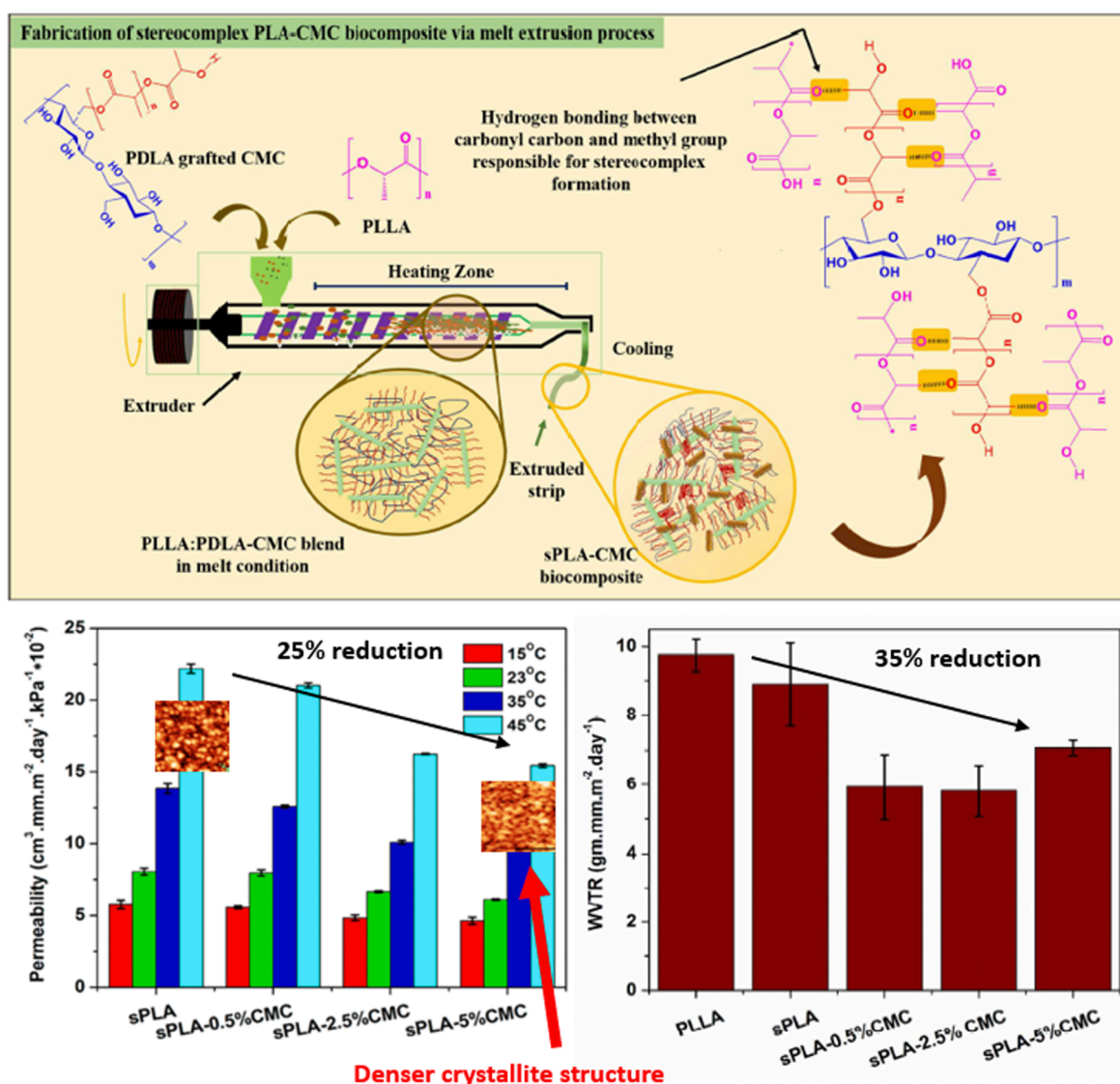


Fig. 10. Influence of stereocomplex crystallite chemistry on the crystallinity and barrier properties of PLLA films [125]. Adapted with permission from ref. [125] from American Chemical Society. Copyright ©2017.

properties. For example, Yuniarto et al. developed a solvent cast PLA film with PEG additives, in which the PEG was incorporated to accelerate the crystal formation in the PLA film matrix [122]. The authors reported a maximum increase (20%) in the oxygen barrier properties with an increase in the degree of crystallinity from 17.71% in the neat PLA to 34.64% in the PLA-PEG films. Some sustainable thermoplastic polyesters, such as PLA, PHA, PGA, and PBS generally possess a higher degree of crystallinity and lower free volume than natural macromolecule derived polymers, such as polysaccharide-based, lipid-based, and protein-based materials [123].

Several studies have also pointed out that the barrier properties of semi-crystalline polymers can be improved through stress-induced crystallization and orientation in the remaining amorphous phase. The orientation of the crystalline structure depends on the strain imposed during polymer processing. Spherulitic morphologies are formed under a low level of strain, while the “shish kebab” structure is generated under a high level of strain (Fig. 9a) [114]. Typically, the “shish kebab” morphologies would increase permeability compared to spherulitic structures, due to the increase in crystallites tortuous arrangement. Bai et al. [124] have proposed a mechanism (Fig. 9b) to overcome the obstacle presented in the “shish kebab” morphologies of PLA. In this study, the authors designed impermeable building block boundaries by interlocking the spherulites with each other by utilizing N,N',N''-tricyclohexyl-1,3,5-benzene-tricarboxylamide (TMC-328) nucleating agents. The resulting interlocked “shish kebab” structure showed a remarkable oxygen permeability reduction by up to 500 times compared to the control PLA, as displayed in Fig. 9c. This is a robust yet straightforward strategy that can be applied to other semi-crystalline biopolymers such as PBS, PHA, and PCL.

In the case of PLA crystallinity specifically, the barrier properties can be subjected to the stereochemistry behavior called “stereocomplex crystallites structure”, which are formed by hydrogen bonding between enantiomeric poly(L-lactic acid) (PLLA) and poly(D-lactic acid) (PDLA) (Fig. 10). Gupta and Katiyar successfully fabricated biocomposites based on PLA and cellulose microcrystal (CMC), which entailed the formation of “stereocomplex crystallites structure” sPLA-CMC from poly(D-lactic acid)-grafted-CMC (PDLA-grafted-CMC) and PLLA polymers to enhance both mechanical and barrier properties of the composite films [125]. The dense structures of the “stereocomplex crystallites” increased the tortuous pathways of gases and slightly reduced oxygen and water vapor permeability by 25% and 35%, respectively. Though the barrier properties were not substantially improved, the work did present a solvent-free and mechanical approach to improve the barrier properties of PLA by utilizing the unique crystalline structure.

Overall, obtaining oriented morphologies to achieve high crystallinity in biodegradable and sustainable polymers is a major challenge, especially for polymers with rich-crystalline phases like PLA or PGA. Maintaining maximum crystalline structure in industrial-scale processes like extrusion, compression molding, or injection molding requires longer annealing or specialized and optimized processing conditions. In order to maximize moisture and oxygen gas barrier properties of renewable packaging coatings and films, future studies should focus on the structure-properties relationship and establish the optimum fabrication process to maintain the highest degree of crystalline structures within the material.

3.4. Renewable polymer blends

Polymer blending is one of the most utilized techniques in the industry as it is a straightforward, cost-effective, and scalable strategy to improve the physical, mechanical, and barrier properties of many biodegradable and renewable polymers. Nevertheless, most polymer blends suffer from lack of interfacial compatibility and viscosity variation resulting in separated morphologies and inferior barrier performances. As mentioned previously, most renewable, and biodegradable blends are immiscible or partially miscible blends that can form particulate (droplets), lamellar (fibrillar), or co-continuous phases (Fig. 5a). One common problem with biodegradable polymer blends is that it is usually challenging to generate blend coatings and films balanced in both barrier and mechanical properties, as such materials also suffer from a lack of appropriate mechanical properties. Different strategies, with various degrees of success, have been employed to improve the interaction between sustainable polymer blends to simultaneously enhance the mechanical and barrier properties. Compatibilization strategies, such as in-situ methods: bridging, chain extension, and reactive compound grafted (maleic anhydride, acrylic anhydride, glycidyl methacrylate) onto the polymer, or ex-situ methods: block co-polymer and graft-copolymer, have been shown to improve the barrier properties of blends [16].

Another type of renewable polymer blend that is commonly applied in the food packaging industry is the emulsion-based blend. Emulsion-based blend coatings and films can be found as protective layers that can prolong the shelf-life of both fresh and processed food products such as fruits and vegetables, cheeses, meat, and bakery products. These emulsion-based coatings typically contain lipid-based materials, providing enhanced barrier performance compared to edible coatings made from one component, including polysaccharide-based [126] or protein-based [127] coatings. Lipid-based materials that originate from bio-resources such as waxes, fatty acids, and vegetable oils are utilized as blends for edible fruits coatings and films because they provide extra resistance to moisture due to the non-polar nature of such materials [128,129]. Since the lipid phase needs to be stabilized in the aqueous phase of the edible coating and film formulations, the emulsification process employing emulsion stabilizing agents is essential. Recent research on emulsion coatings and films has investigated a wide variety of food-grade emulsifying agents that can stabilize the emulsion and increase the distribution of lipid particles. The use of either classical emulsion stabilizers like Tween 80, lecithin, polysorbates, mono and di-glyceride ester [130] and Pickering emulsion stabilizers such as surface-modified starch, chitosan, hydroxyapatite, and nanocellulose [131] displayed appealing results. However, there are still some concerns regarding the longevity of stability and safety of these materials that need to be addressed before commercial deployment. The concern is especially valid for nanomaterial Pickering emulsifiers as their safety is not widely established.

3.5. Multilayer assembly

Another scalable technology commonly used in food packaging applications is multilayer assembly. Compared to polymer blends,

multilayer assembly can achieve much higher moisture and/or oxygen barrier properties in food packaging [132]. The various layers of polymer stacking on each other in multilayer coatings and films resemble the laminar structure in polymer blending, which would theoretically provide the most optimal moisture and oxygen gas barrier (Fig. 5b). Due to the superior structures, they constitute and barrier properties they offer, multilayer assembly films are extensively utilized in various packaging applications, such as beverage, food, cosmetics, pharmaceutical, and electronics. Multilayer assembled films, sheets and films dominate the food packaging industry by constituting about 73.5 % of the total packaging used (Fig. 11)[133,134]. However, the performance of the multilayer coatings and films is also highly dependent on the adhesive strength and chemical interaction between each level of layers. Similar to polymer blends, the bio-originated and biodegradable polymer used in multilayer assembly must have a certain degree of compatibility, which can either be achieved through the selection of materials or by using compatibilization techniques.

Conventional multilayer films typically use two to five different polymers that can be fabricated into multilayer assembly of two to eleven layers arrangement and can further be increased up to thousands of layers with microscale thickness by micro co-extrusion process [135,136]. Common methods to manufacture biodegradable multilayers coatings and films include either dry processes such as co-extrusion/ lamination/compression molding or wet process such as solvent-coating/ aqueous dispersion/ vacuum coating [132].

Standard multilayer films are typically comprised of the outer layer, adhesion (tie), barrier and inner seal layer, as shown in Fig. 12. For enhanced barrier packaging application, the two most important layers are the adhesion and the barrier layers. The adhesion layers are incorporated to allow different barrier layers to adhere and interact with each other within the multiphase polymer system, as well as to prevent delamination [137]. For renewable materials, the interactions between polymers are mostly poor due to the chemical incompatibility and immiscible coupling chain diffusion. Thus, the adhesion layers are made up of modified polymers with enhanced interfacial affinity with barrier layers. Examples of modifications on the adhesion layer include: grafted co-polymers [138], maleic anhydride (MA) grafted polymers [99,139], and reactive surface blends [140]. On the other hand, the barrier layer is the protective layer that provides improved permeability properties in the packaging material. Some candidates for a biodegradable polymer that can act as barrier components for multilayer assembly are PVOH, PGA, PHB and TPS for oxygen barrier properties [81] and PGA, PLA, and hydrophobic protein biopolymers (zein, gelatin, gluten) for moisture barrier properties [57]. One of the most appealing polymers used for multilayer coatings and films is PVOH, which can act as additional supportive layer that provides excellent oxygen barrier properties in structures like tri-layer PHA/PVOH/PHA [139], tri-layer LDPE/PVOH-TPS blend/LDPE [141], bilayer silk fibroin/PVOH [13], and bilayer PVOH/CNC films [142]. To further enhance the moisture and oxygen barrier properties, the addition of micro and nanofillers into the barrier layers such as nano-clay, nanocellulose, graphene, and metal oxide can also be employed [142–144].

Overall, the technology of multilayer coatings and films can be applied to vastly upgrade the overall barrier properties of renewable and biodegradable materials by providing extra barrier layers tailored to moisture/oxygen permeability [37]. However, one crucial problem that these industrial multilayer coatings and films need to address is that all components and layers in the assembly, including the adhesive layer, need to satisfy the sustainability requirement to ensure that the whole structure provides the desired environmental

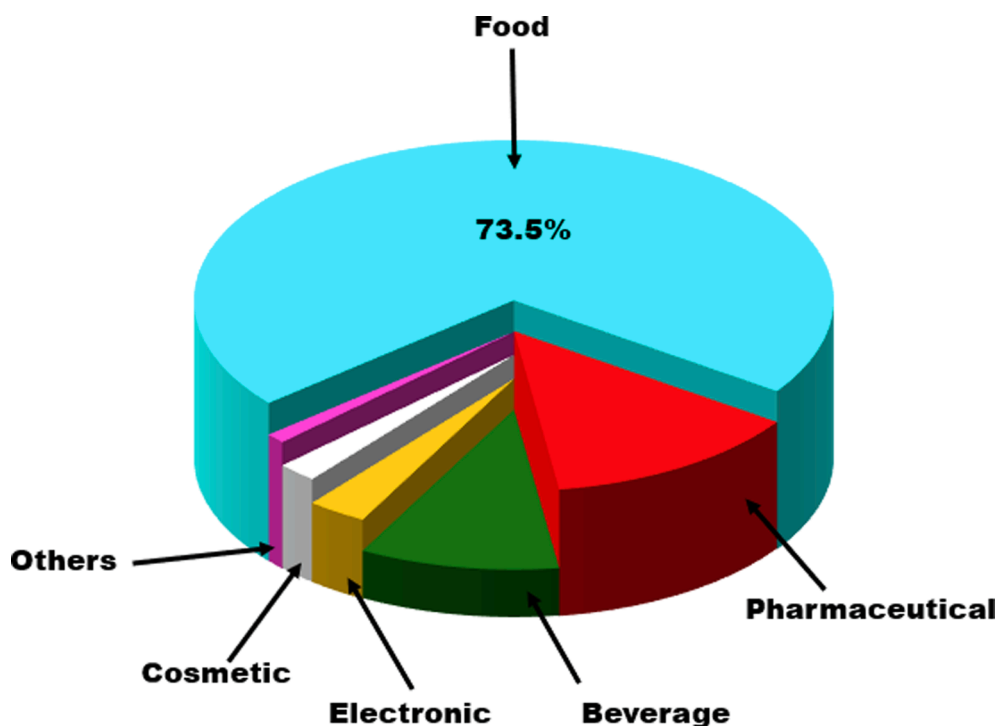


Fig. 11. A graphical breakdown of multilayer technology coatings and films by industrial applications in 2019. Data obtained from ref. [133,134].

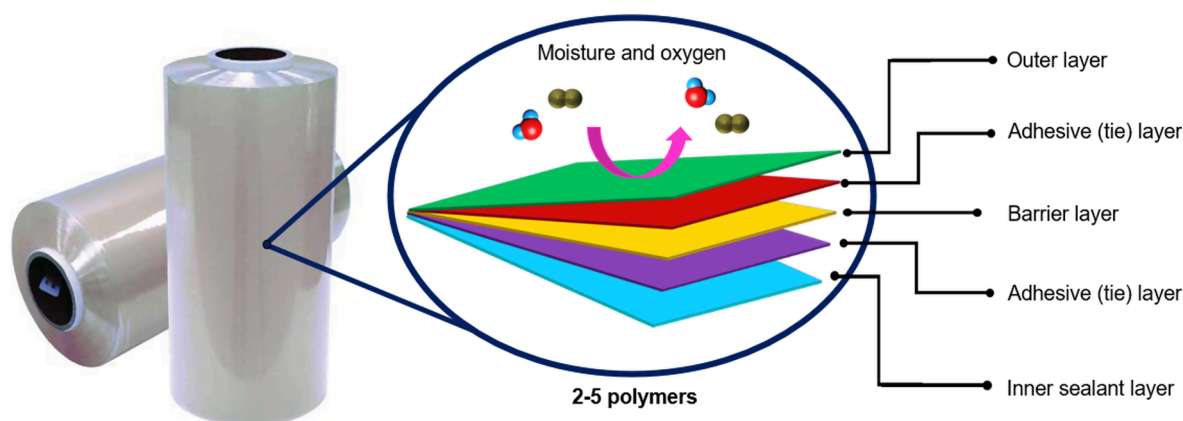


Fig. 12. Layer-by-layer arrangement of a standard multilayer film system.

attribute. Thus, more research on the biocompatibility and variation of existing polymer multilayer combinations should be carried out to help fabricate high barrier coatings and films while maintaining the biodegradability and recycling capacities of the plastics.

3.6. (Nano)Composites and other additives

The incorporation of nanofillers and additives into packaging coatings and film can tremendously enhance the overall barrier performance by filling the free volume and thus reducing the porosity as well as by providing a tortuous pathway for the diffusion of permeants. In the last few decades, the use of nanoparticles and nanocomposites as novel materials has been extensively studied, and many researches have demonstrated successful attempts of improving the barrier properties of polymer matrices with the incorporation of a minimal amount (<10 wt.%) of these nano-sized fillers [90,145,146].

The selection of impermeable nanoparticles to moisture, oxygen, and other gases is crucial to inhibit the diffusion path of such permeants in the polymer matrices. Moreover, obtaining high barrier performance often requires the nanofillers to have a uniform and homogenous dispersion inside the composite systems. As a result, it is not rare to see nanoparticles to aggregate and agglomerate due to thermodynamic reasons and poor chemical incompatibility with the polymer matrices, which can ultimately cause detrimental effects to the polymer such as brittleness, weakened mechanical properties, reduced transparency, as well as diminished barrier performances [61,147]. However, these problems can be mitigated by employing appropriate chemical modifications of the fillers or the polymers, using surfactants and/or compatibilizers, or by exploiting advanced processing techniques, such as the use of ball milling, high shear mixing, and ultrasonic treatment [148,149]. Some of the vital nanocomposite systems that affect the overall barrier performance, as illustrated in Fig. 13, can be categorized as the followings:

- (1) Nano-clay-based: Clay nanoparticles typically take the form of nanoplatelets, which makes them excellent candidates as reinforcing materials that inhibit the permeation of moisture/oxygen through coating and film membranes.
- (2) Cellulose-based: Cellulose-derived nanoparticles have shown to possess natural excellent oxygen barrier properties due to the poor polarity interaction with O_2 molecules. Another advantage of utilizing cellulose-based nanoparticles is that these nanoparticles can easily be tailored by surface modification to be incorporated into the appropriate food packaging system.
- (3) Carbon-based: Carbon and biocarbon-based materials, like graphene oxide (GO), graphene nanoplatelets (GNP), and carbon nanotube (CNT), are often incorporated in packaging films as high barrier and antibacterial additives [150]. Due to their high surface area and aspect ratio, only a small concentration of carbon-based nanomaterials is required in packaging materials to achieve superior barrier properties compared to other types of nanofillers. Carbon-based nanoparticles can also be an extremely effective sources of materials to enhance the mechanical and thermal properties of packaging films [151].
- (4) Others: High barrier coatings and films can also utilize other sources of bio-based nanofillers such as natural short and long fibers (hemp, flax, jute, bamboo) [152,153], lignin nanoparticles (LNP) [154] and solid lipid nanoparticles (SLNP) [155]. Additionally, inorganic metal-based nanoparticles, such as those of zinc oxide and silica, can also be added to a biodegradable polymer systems to enhance the barrier properties, as well as to provide antimicrobial activities of the coatings/films [156].

Exemplary nanofillers and their improvement on the barrier properties is summarized in Fi. 2.9, and discussions are provided in the subsequent sections.

4. Recent development of enhanced barrier performance of renewable polymers in coatings and films

4.1. Barrier improvement using compatibilizers

Improving the barrier properties of various multiphase and multicomponent biodegradable films has been a topic of high research

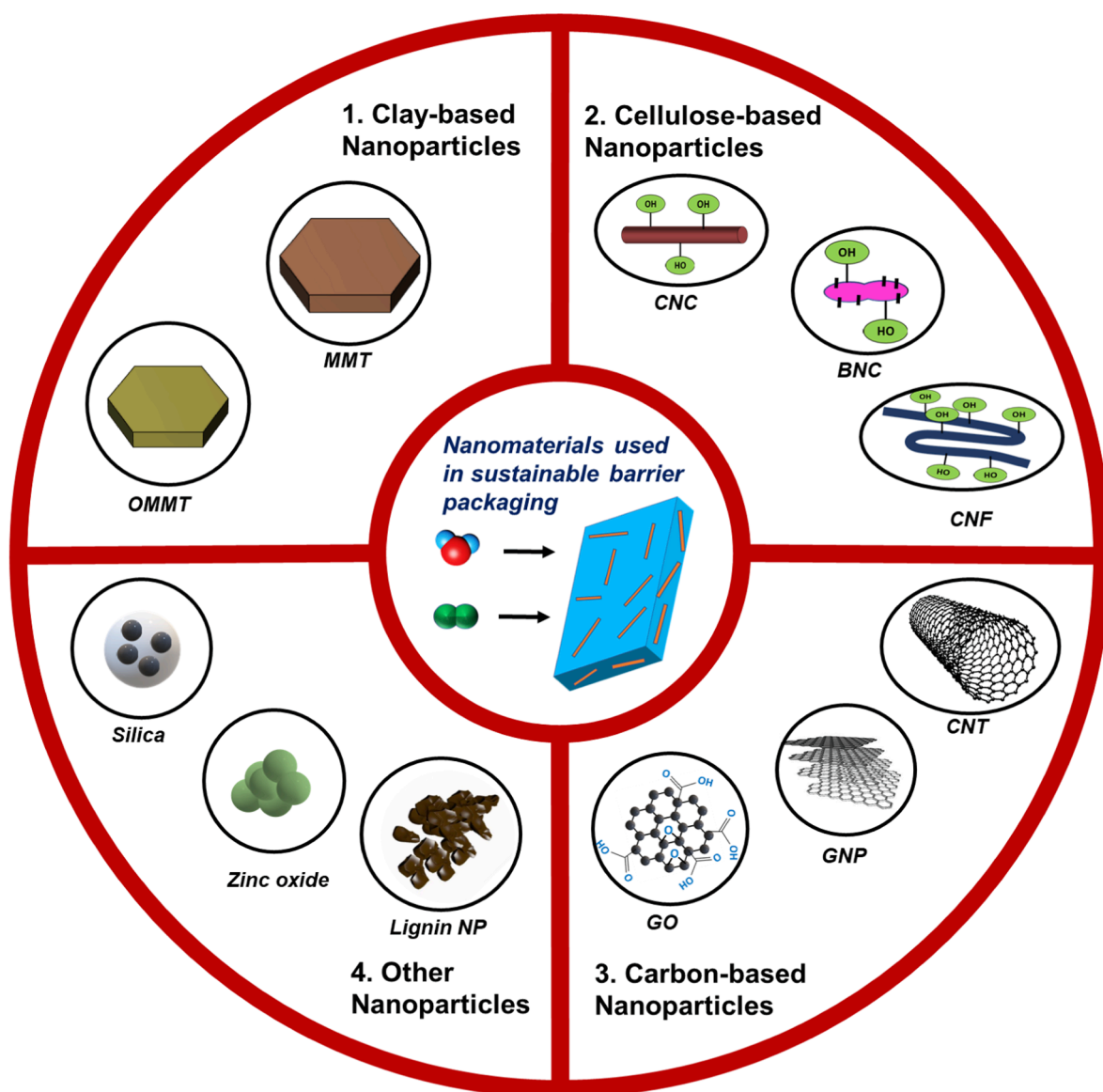


Fig. 13. Different types of nanoparticles used in sustainable packaging coatings and films.

interest to ensure their success in practical applications. One of the proposed variables to control the permeation of moisture/gas through biodegradable films and coatings is to use compatibilizers in the multiphase and multicomponent polymer system. It is believed that better interaction and dispersion of immiscible blends enhanced by compatibilizer can help the structure improve the overall mechanical and barrier properties [67,157]. There have been many promising studies on compatibilizers to improve the film's barrier properties for edible packaging films that have been reported elsewhere. However, most of these reviews focus on the influence of compatibilizer on mechanical and thermal properties instead [55,158,159]. To the best of our knowledge, there has been no specific reviews discussing the improvement in barrier properties influenced by compatibilizers. Overall, compatibilizers can be separated into two main categories [160]:

- (1) ***In-situ compatibilization:*** Reactive functional groups (i.e., acrylate, anhydride, isocyanate, epoxy) are used to improve compatibility and increase the miscibility of multicomponent systems. These reactive groups can react or form interlocking molecular bonding with other functional groups (i.e., hydroxyl, carboxyl, ester) of biodegradable polymers, which ultimately increase the interfacial interactions between components in a blend or multilayer structure. In addition, these reactive functional compatibilizers can be incorporated into the biodegradable multicomponent system through reactive extrusion.
- (2) ***Ex-situ compatibilization:*** Premade copolymers are used to improve the compatibility between the phases in multicomponent systems. The compatibilizing copolymers can be either block-copolymers (building block of two or more polymers are connected to form a unit) or graft-copolymer (polymers are built and branched out from the backbone of the other polymer chain).

Table 3

Improvement in barrier properties using compatibilizer.

	Compatibilizers	Multicomponent system	OP/OTR improvement (%)	WVP/WVTR improvement (%)	Ref
In-situ	Ethylene-methylacrylate-glycidyl methacrylate (EMA-GMA)	PGA/PBAT blend (50/50 wt.%)	17.6% (23°C, 50% RH)	47% (23°C, 65% RH)	[49]
	Nisin-ethylenediaminetetraacetic (Nisin-EDTA)	PBAT/TPS blend (60/40 wt.%)	25% (23°C, 50% RH)	15% (25°C, 50% RH)	[161]
	Epoxy group AX8900	PPC/PBSA blend (70/30 wt.%)	-	15% (23°C, 90% RH)	[162]
	Chloropropyl trimethoxysilane (CPMS) coupling agent	TPS/PLA blend (50/50 wt.%)	76.5% (23°C, 0% RH)	-	[163]
	Maleic anhydride (MA)	TPS/PLA multilayers	96% (23°C, 50% RH)	80% (23°C, 50% RH)	[99]
Ex-situ	Poly lactide-graft-glycidyl methacrylate (PLA-g- PLA-g-GMA) copolymer	PLA/ 1 wt.% lignin composite	51% (23°C, 50% RH)	-	[164]
	Poly lactide-graft-poly(ethylene glycol) (PLA-g-PEGMA) methyl ether methacrylate	PLA/ 1 wt.% lignin composite	81% (23°C, 30% RH)	-	[164]
	Starch-graft-poly(lactic acid) (St-g-PLA) copolymer	PLA/TPS blend (30/70 wt.%)	42.9% (23°C, 30% RH)	96% (23°C, 30% RH)	[165]

One block from premade copolymer is miscible with a material in the multicomponent system. In contrast, the other block of copolymer is miscible with the other component, resulting in partial miscibility in a blend, multilayer, or composite system.

Table 3 presents some research on the effects of compatibilization techniques on the improvement of barrier properties within sustainable multiphase and multicomponent systems. Overall, the use of in-situ reactive compatibilizers has shown potentials to effectively help multicomponent systems, especially polymer blends, to help maintain the low-level moisture/gas permeability. On the other hand, studies on the effects of ex-situ compatibilization techniques on the overall barrier properties of polymer blends are still not thoroughly investigated.

4.2. Blends vs. Multilayer structures

Different biodegradable polymers have been combined, either in the form of polymer blends or multilayer films, to improve the packaging materials' overall intrinsic properties. To achieve high moisture/oxygen barrier for multiphase and multicomponent packaging films, at least one polymer component with inherently excellent barrier properties must be incorporated into these systems. Nowadays, a combination of both blends and multilayer structures is of interest for packaging applications, and the resulted films are

Table 4

Comparison of oxygen and water vapor barrier improvements in blends versus. multilayer assembly.

Polymer matrices	Blend films				Multilayer films			
	Processing detail	OP/OTR improvement (%)	WVP/WVTR improvement (%)	Ref	Processing detail	OTR/OP improvement (%)	WVTR/ WVP improvement (%)	Ref
TPS/PLA	- Melt extrusion and compression molding- Compatibilized by epoxidized oil- PLA/TPS ratio (80/20 %)	0% (25°C, 50% RH)	90% (23°C, 50% RH)	[171]	- Compression molding and dip coating.- TPS layer is compatibilized with MA-Trilayers of PLA/TPS/PLA	96% (23°C, 50% RH)	80% (23°C, 50% RH)	[99]
TPS/PBAT	-Blown film extrusion- TPS/PBAT ratio (50/50)- TPS is modified with polyurethane prepolymer	55% (25°C, 40% RH)	-	[172]	- Compression molding and dip coating.- TPS layer is compatibilized with MA- Trilayers of PBAT/TPS/PBAT	74% (25°C, 50% RH)	87% (25°C, 50% RH)	[140]
PLA/PBS	- Solvent casting- PLA/ PBS ratio (80/20 %)	32%(25°C, 53% RH)	18% (25°C, 53% RH)	[173]	- Multilayer co-extrusion- PLA/PBS ratio (80/20 %)	30%(25°C, 50% RH)	40%(25°C, 50% RH)	[174]
PLA/Chitosan	- Solvent casting- PLA/ Chitosan ratio (30/70%)	-	25% (25°C, 90% RH)	[175]	- Layer-by layer assembly- Chitosan layers reinforced with CNC	-	30%(23°C, 85% RH)	[176]

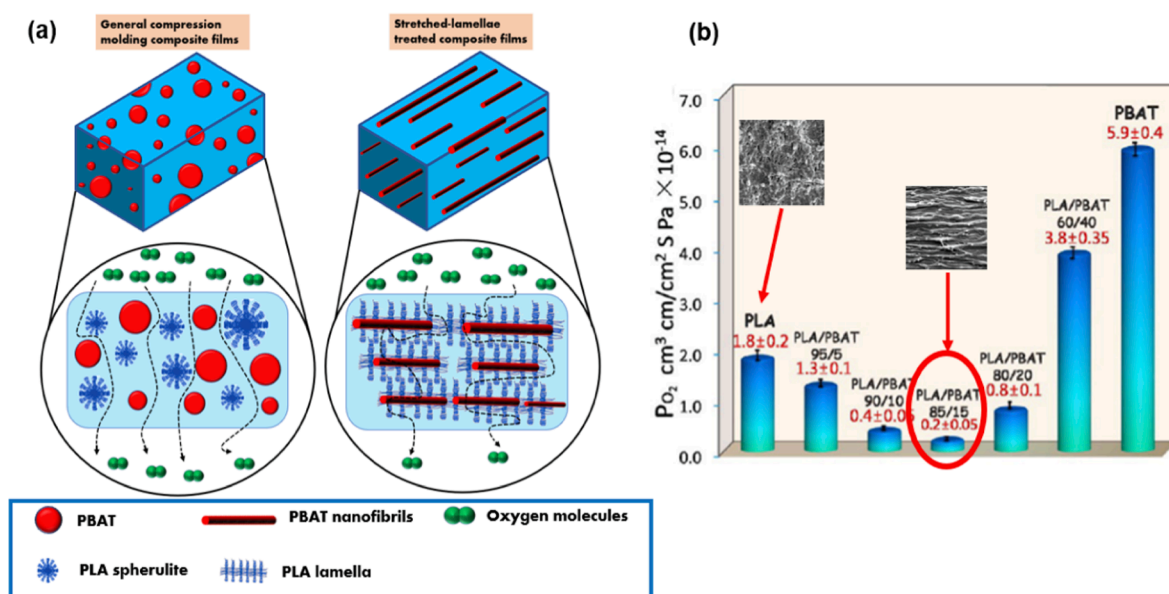


Fig. 14. (a) Schematic of super-barrier films made from oriented lamellae PLA and nanofibrils PBAT. (b) Oxygen permeability improvement of the lamellae PLA/PBAT blends. Adapted with permission from ref. [65] from American Chemical Society. Copyright ©2016.

expected to provide excellent barrier properties while maintaining good flexibility and affordability. Table 4 summarized and compared the improvement in the barrier characteristics between the multiphase blends and the multilayer assembly of various renewable and biodegradable multicomponent polymer systems. However, it must be noted that relevant research on completely biodegradable protective films with high barrier properties is limited, as most multilayer systems still require traditional non-biodegradable materials such as PE [166–168] or EVOH [68,169,170] to maintain good flexibility and barrier capabilities. Based on the results provided by Table 4, the multilayer systems typically possess superior improvement in oxygen and water vapor barrier compared to those of polymer blend films. This was attributed to the hierarchical barrier functionalities of multilayer structures, in which each layer will exert and stack their maximum protective effects together [132]. On the other hand, polymer blends are only expected to show mechanical and barrier properties bounded by the natural characteristics of the original materials [55].

The microscopic morphologies of the blend are also critical factors that can lead to the change in barrier properties. In some particular cases, the orientations and morphologies of the polymer chains can significantly improve the moisture/oxygen barrier properties of the materials. Recently, Zhou et al. reported that reduction in oxygen permeability can be achieved in a blend with the appropriate change in morphological orientations [65]. The researchers successfully generated a blend of PLA/PBAT, in which the PBAT has an oriented parallel lamellae structure inside shish-kebab crystallized PLA films, as seen in Fig.14. The lowest oxygen permeability for this PLA/PBAT blend was achieved at 15 wt.% PBAT, which was 9 times and 30 times lower than PLA and PBAT, respectively. This proposed fabrication method of lamellae flow blends introduces a robust scalable process that can construct high barrier film structures without chemical compatibilizers.

4.3. Nanocomposites films

Fillers, especially nanoparticles, provide an excellent moisture/oxygen barrier for food packaging and coating applications attributed to their high crystallinity and high specific area/aspect ratio. The incorporation of nanoparticles as fillers is frequently

Table 5
Oxygen/water vapor improvements of nanocomposite films.

Nanocomposite system	Polymer matrices	Nanofillers	Fabrication method	OTR/OP improvement (%)	WVTR/WVP improvement (%)	Ref
Clay-based nanocomposites	PCL	OMMT ($C_{19}H_{42}N_2Cl$)	Melt extrusion	-	57.1% at 3 wt.%(37°C, 20% RH)	[182]
	PLA	OMMT (C30B)	Melt blending	46% at 3 wt.%(38°C, 90% RH)	-	[183]
	PHBV	OMMT (C30B)	Melt extrusion	9.6% at 2.5 wt.%(23°C, 0% RH)	No improvement	[184]
	PHBV/PBAT	OMMT (Nanomer® I.30T)	Cast extrusion	78.9% at 1.2 wt.%(23°C, 0% RH)	69.5% at 1.2 wt.%(37.8 °C, 100% RH)	[185]
Cellulose-based nanocomposites	Chitosan	CNC	Cast extrusion	-	35% at 10 wt.%(25°C, 60% RH)	[212]
	PLA	CNC	Melt extrusion	40% at 1 wt.%(23°C, 0% RH)	75% at 1 wt.%(23°C, 85% RH)	[192]
	Chitosan	CNF	Solvent casting	-	35% at 5 wt.%(25°C, 60% RH)	[213]
	PLA	CNF+OMMT (C30B)	Solvent casting	90.2% at 5 wt.%(23°C, 50% RH)	76% at 5 wt.%(23°C, 50% RH)	[193]
	PVA (PVOH)	Modified BNC-AgNP	Solvent casting	25% at 5 wt.%(25°C, 50% RH)	90.2% at 5 wt.%(25°C, 50% RH)	[214]
Carbon-based nanocomposite	PLA	GO-CNT hybrid	Solvent casting	66.6% at 0.4 wt.%(23°C, 50% RH)	-	[215]
	PLA	GNP	Solvent casting	68 % at 0.4 wt.%(23°C)	-	
	PLA	Reduced GO (rGO)	Compression molding	99.0 % at 1.01 mg/cm ² (23°C, 50% RH)	87.6 % at 1.01 mg/cm ² (38°C, 90% RH)	[199]
	PCL	CNT	Solvent casting	58 % at 1 wt.%(23°C, 80% RH)	-	[216]
	PHBV	CNT	Solvent casting	62 % at 1 wt.%(23°C, 80% RH)	-	[216]
Other	Gelatin/Chitosan	SiO ₂	Solvent casting	-	80% at 10 wt.%(37°C, 20% RH)	[207]
	PLA	SiO ₂	Melt extrusion	64% at 30 wt.%(23°C, 65% RH)	-	[210]
	TPS	CaCO ₃	Solvent casting	-	70% at 0.06 wt.%(25°C, 50% RH)	[208]
	PLA	ZnO	Solvent casting	33.5% at 9 wt.%(25°C, 50% RH)	39.1% at 9 wt.%(25°C, 50% RH)	[217]
	TPS	ZnO	Solvent casting	40% at 5 wt.%(20°C, 50% RH)	-	[209]
	Gelatin	ZnO	Solvent casting	36% at 9 wt.%(20°C, 50% RH)	-	[209]
	Chitosan	ZnO-loaded gallic acid	Solvent casting	41% at 3.5 wt.%(25°C, 50% RH)	87% at 3.5 wt.%(25°C, 50% RH)	[218]

utilized in packaging applications. Their reinforcing capability usually provides extra mechanical strength (e.g., tear strength, puncture resistance, stiffness, etc.) to the functioning films and coatings [177,178]. In addition, some nanofillers, such as silver [179] and zinc oxide [180], can further provide antibacterial properties. The enhanced barrier properties of bio-nanocomposites generated from various biodegradable polymers and sustainable fillers are summarized in Table 5.

4.3.1. Nano-clay-based nanocomposites

Nano-clays are appealing nanomaterials because they are nature-derived, abundant, low cost, safe, etc. [178,181]. One of the most widely used nano-clays for food packaging and coatings application is montmorillonite (MMT) layered silica platelets with particle thickness of 1–2 nm and a high surface area of more than 700 m²/g [178]. Due to the high surface area and aspect ratio, host polymer matrices can achieve high barrier properties with a low loading percentage of well-dispersed nano-clay particles. Since most of the nano-clays are in the form of nanoplatelets, these nanoparticles can provide a complicated system of tortuous pathways to reduce the permeation of moisture/oxygen permeants. It is widely reported that MMT can easily be dispersed in either exfoliated or intercalated structures in hydrophilic polymer films, such as polysaccharide-based or protein-based films using typical solvent or thermo-mechanical polymer processing techniques. On the other hand, MMT is not suitable as a barrier nanomaterial for hydrophobic polymers, including several hydrophobic polyesters, as it forms aggregates and agglomerates due to low compatibility. For MMT to be utilized as an effective functional filler in hydrophobic biodegradable polyesters, such as PCL [182], PLA [183], PHBV [184,185], and PBAT [185], the hydrophilic silicate surface must be modified into organophilic OMMT to reduce the surface energy and improve its interaction with these hydrophobic polymers [181].

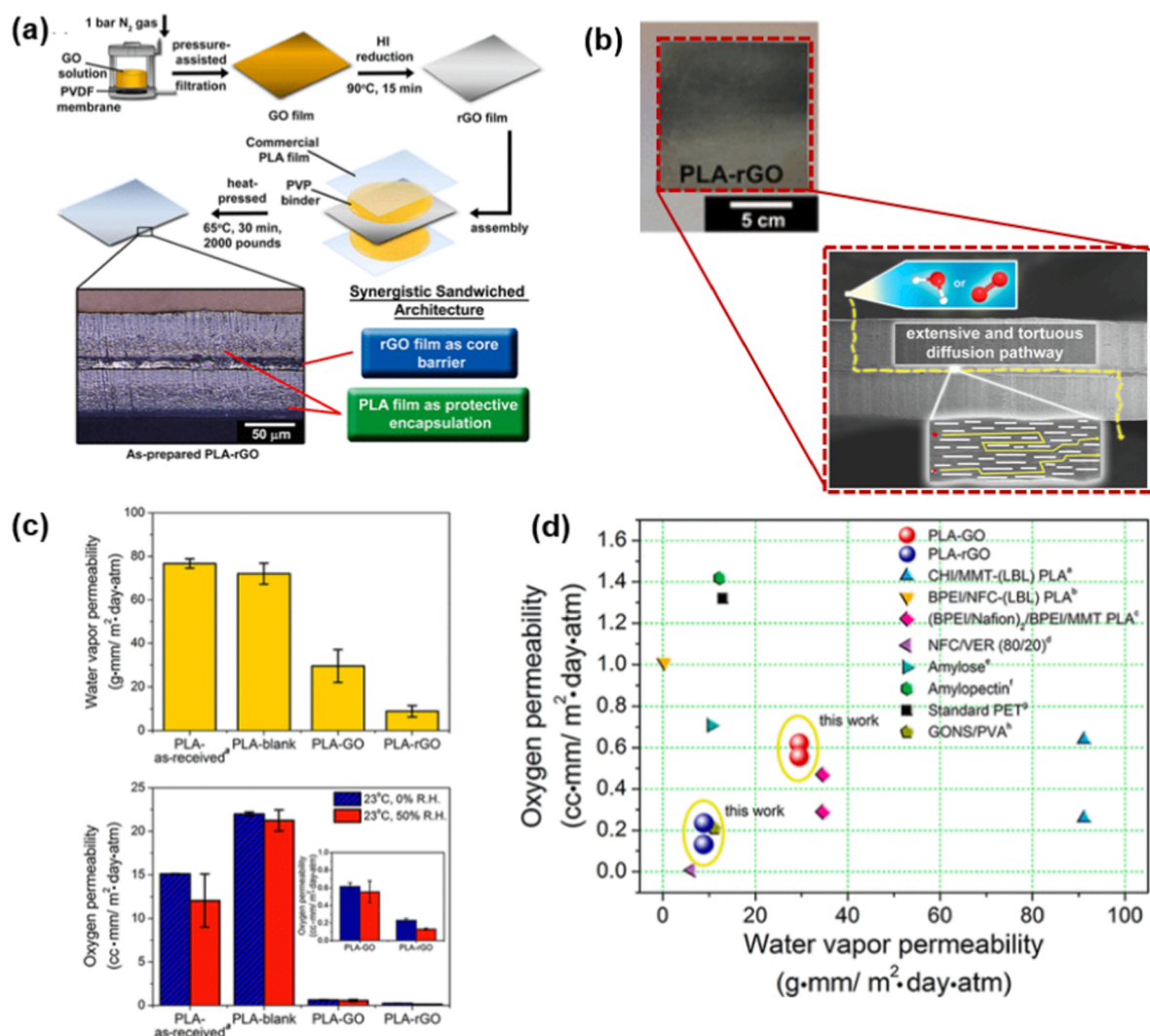


Fig. 15. (a) Procedure to prepare PLA-rGO composite. (b) Mechanism of enhanced barrier properties indicated through SEM of the PLA-rGO composite film (c) WVP and OP improvement of the PLA-rGO composites. (d) Comparison of WVP and OP of PLA-rGO with other established packaging systems. Adapted with permission from ref. [199] from American Chemical Society. Copyright ©2016.

It is worth mentioning that the barrier properties nano-clay offers can be highly dependent on the nature of the host polymer and can be overshadowed by the barrier properties of the polymer matrices. Research by Crétois et al. shed light on this topic by studying the influence of OMMT in the PHBV/OMMT nanocomposite in terms of mechanical and barrier properties [184]. The results showed that the moisture permeability of the PHBV/OMMT nanocomposites was not affected by the nano-clay content in the matrices, even at high concentration loading (>10 wt.%). Therefore, the authors suggested that the high crystallinity contributed by PHBV already has higher barrier properties than the nano-clay, which makes it redundant to add nano-clay as a moisture barrier material.

4.3.2. Cellulose-based nanocomposites

Cellulose-based nanomaterials (CNC, CNF, BNC) are appealing bio-resourced nanofillers that can contribute to high barrier properties when incorporated into the proper polymer coatings and films. This is attributed to the little to no amorphous region in these nanocellulose structures that leave predominantly crystalline regions [186,187]. Moreover, the natural abundance of cellulose make them enticing to be capitalized as sustainable nanomaterial feedstock [188]. These nanofillers can act as nucleation agents inducing high crystallinity in the polymer matrix, and they might also interact with the polymer matrix, such as forming hydrogen bonding and inter-crosslinking to provide optimal properties [51,189]. Moreover, the high aspect ratio of nanocellulose, the abundance and safe-for-consumption quality make them very attractive for barrier materials in food packaging application. Short and long cellulosic fibers (e.g., hemp, flax, jute) have shown appealing potential as barrier fillers due to their abundance, as well as providing lamellar-orientated fillers which further reduce permeability in packaging composites [190]. In the last few decades, research on nanocellulose use in nanocomposite materials for packaging coatings and films has been extensively explored and steadily increasing [191]. However, due to the natural hydrophilicity, these cellulose-based fillers can only improve the water vapor barrier of polymers by a modest amount compared to oxygen permeability, and the barrier effects are typically weakened at higher relative humidity [192].

In some cases, incorporating additional nanofillers is required along with nanocellulose fillers to enhance the moisture/oxygen barrier properties. For example, a study by Trifol et al., found that the water vapor barrier of PLA did not significantly improve by adding just CNF. However, adding extra Nano-clay C30B as hybrid nanocomposite helped improve the PLA nanocomposite films by 90% in WVP [193].

To solve the limitation of nanocellulose on gas barrier properties, chemical modification and surface grafting by utilizing the –OH group is a widely employed strategy that aims to decrease the polarity of nanocellulose and enhance its compatibility with hydrophobic polymer matrices. These nanocellulose surface modification chemistries reported for food coatings and film applications include but are not limited to succinylation [194,195], phosphorylation [196], oxidation [197], and enzymatic esterification [198]. Overall, the development of green chemistry and processing methods to fabricate modified bio-nanocomposites are expected to significantly improve the dispersion of fillers inside polymer matrices, thus, enhancing the overall barrier properties of materials in packaging.

4.3.3. Carbon-based nanocomposites

Carbon-based nanomaterials, in the form of graphene oxide (GO), graphene nanoplatelets (GNP), biocarbon, and carbon nanotube (CNT), have been investigated in various systems of sustainable polymers as impermeable fillers to improve the physical, thermal, morphological, and gas permeation properties. Generally, carbon-based nanoparticles have shown to provide effective improvement in the mechanical and barrier properties of films at small concentration of carbon fillers (< 1 wt.%), as compared to clay-based and cellulose-based nanocomposites, where the fillers are usually loaded at 5-10 wt.% to achieve the same effectiveness. These high potency in term of barrier properties provided by nanocarbons are attributed to their high aspect ratios and large surface areas, which

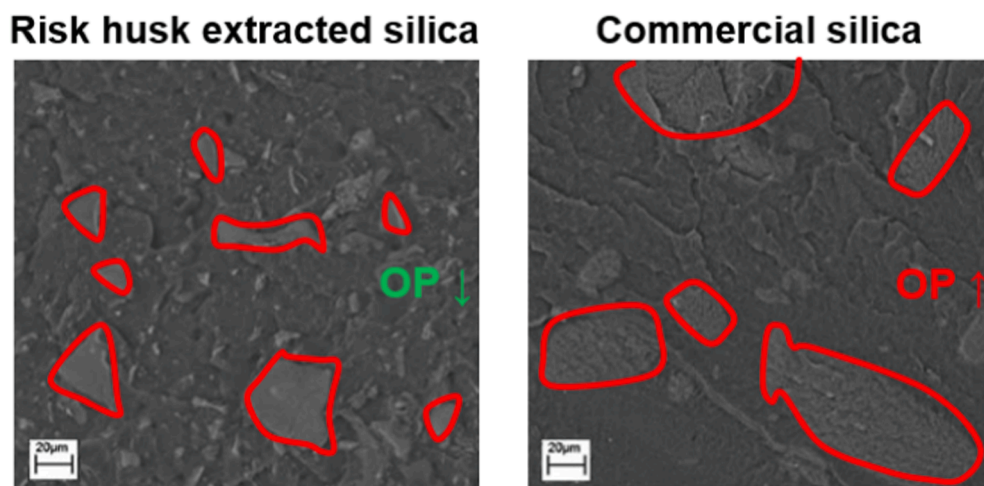


Fig. 16. Dispersion of silica nanoparticles for different source in gelatin/silica films. Adapted with permission from ref.[210] from Royal society of chemistry.

allows fine dispersions in the composites [199,200]. The excessive loading of carbon-based nanoparticles are prone to agglomeration and phase separation due to stronger intermolecular Van der Waals interactions that cause detrimental effects in the gas barrier properties for packaging films [201]. For instance, a study by Sanchez-Garcia et al. investigated the effects of increasing the loading levels of carbon nanotube (CNT) in both PCL and PHBV biocomposites [202]. The results demonstrated that the OP of both PCL-CNT and PHBV-CNT composite films were at their lowest at 1 wt.% CNT loading, but the OP reverted to the original values of non-filled matrices at 5 wt.% CNT concentrations, showing non-effective oxygen barrier performance due to agglomeration. Overall, the outstanding impermeable properties of carbon-based nanomaterials seem to work best only at controlled low concentrations, and rarely have direct correlations to the filler loading levels, as observed in clay or cellulose-based nanocomposites.

To enhance the water vapor/oxygen barrier properties of carbon-based nanocomposites, various effective strategies were employed that typically help with the dispersion and orientation of nanocarbons in the polymer matrices, with prime examples being chemical modification and arrangement of particle lamellae structures. Similar to surface modification of nanocellulose, nanocarbon structures can also be changed to enhance the interactions between fillers and polymer matrices. For example, Zhou et al. grafted superhydrophobic moiety (octadecyl amine (ODA)) onto the surface of GO to promote the hydrophobic interaction between PLA and GO nanofillers [203]. The results showed that the modified GO underwent superhydrophobic transformation to become GO-ODA, reflected in the WCA increase from 21° to 164° . This has significantly contributed to the reduction of WVP as explained by two mechanisms: (1) GO became water-repellant and block more water vapor penetrating into the film, and (2) the tortuous pathways for permeants in the nanocomposites were enhanced by the improved homogeneous dispersion of GO and improved compatibilities between hydrophobic PLA and GO-ODA. In a different study by Goh et al, the promotion of water vapor and oxygen barrier of commercial PLA nanocomposite were addressed by a different sandwich-architecture design, which introduced reduced GO (rGO) between bilayers of PLA, rather than imbuing rGO directly in the PLA [199]. The OP and WVP values have reduced by 99% and 87.6%, respectively. This impressive improvement in barrier properties of PLA-based composites was attributed to the compact lamellar arrangement of rGO, which enhanced the extensive and tortuous network of permeants' travel pathway, as indicated by Fig. 15.

4.3.4. Other nanocomposite fillers

Researches on other nanomaterials for renewable coatings and film composite applications that provide barrier property improvement includes natural fibers [153,204], lignin nanoparticles (LNP) [205,206], silica (SiO_2) [207], calcium carbonate (CaCO_3) [208], and metal zinc oxide (ZnO) [180,209]. Table 5 shows various examples of these nanoparticles and the achieved enhanced barrier attribute resulting from filler incorporation. However, it should be noted that the effects on the barrier properties provided by these nanoparticles are not as straightforward as those of mechanical and thermal properties, and there are usually some drawbacks to

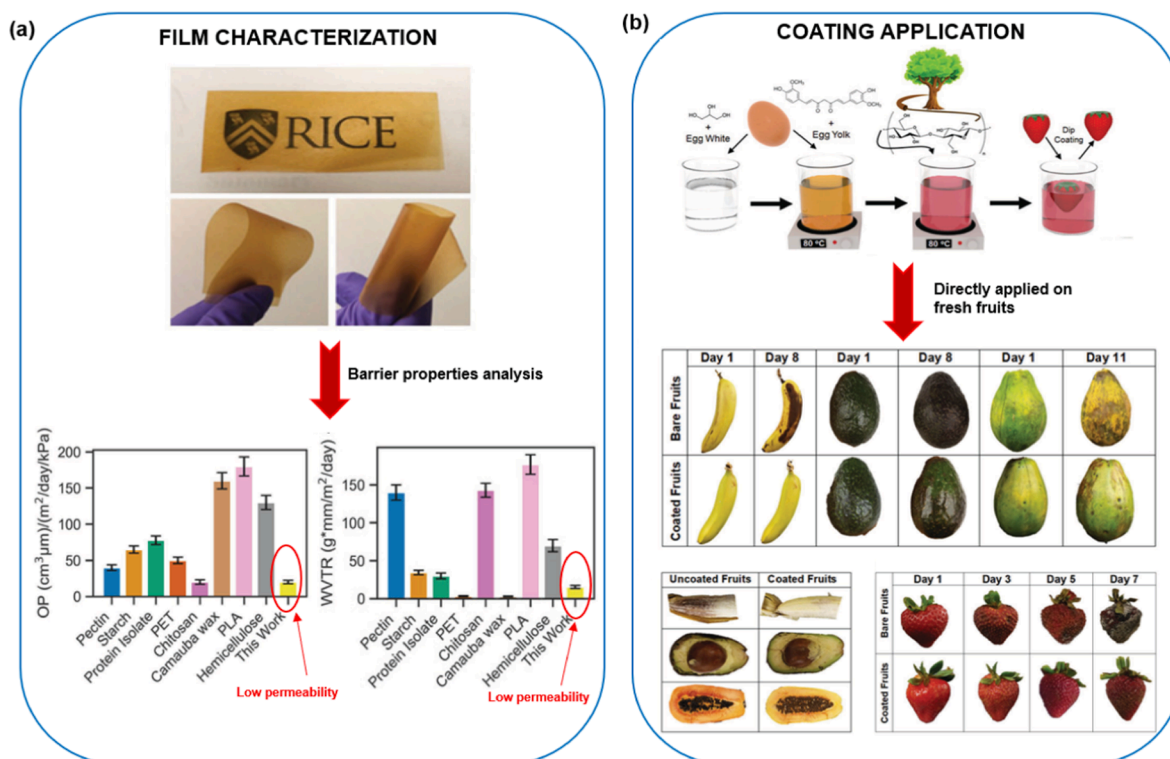


Fig. 17. (a) The film of CNC reinforced poly(albumen) nanocomposite films and their improvement in moisture and oxygen barrier properties. (b) Schematic of the fabrication process of poly(albumen) nanocomposite dipping solution and their effects on different fruit ripeness. Adapted with permission from ref. [224] from Wiley Online Library. Copyright ©2020.

the nanocomposites associated with them. Batra et al. [207] reported the inclusion of silica nanoparticles in their gelatin/chitosan/and silica bio-nanocomposite films. The study results showed a WVP reduction by 80% with the incorporation of 10 wt.% silica nanoparticles [207]. However, the authors also reported the incorporation of silica nanoparticles increased the moisture absorption and entrapped water vapor inside the gelatin channels, likely because of the blockage caused by the nanoparticles.

Additionally, the barrier properties of polymer matrices can also be affected by the source of silica, as suggested in the work of Battegazzore et al. on the comparison between OP values of commercial silica and rice husk extracted silica in PLA films [210]. The authors observed an improvement of 64% OP reduction at 30 wt.% rice husk extracted silica, while the OP trend for commercial silica was completely opposite at an increase of 1567% at the same loading of 30 wt.%. The observation resulted from the difference in aggregation shapes and sizes of the silica formed in the PLA matrices (Fig. 16). The authors proposed that aggregation caused by commercial silica may have caused internal defects in the film's structure and thus, increased the OP values. Sun et al. investigated the mechanical properties and moisture barrier permeability of corn TPS/CaCO₃ and found that the nanocomposites' WVP has significantly reduced by up to 70% with 0.06 wt.% CaCO₃ loading [208]. Nevertheless, the surface morphology of the nanocomposite films displayed deep cracks with an increase in the CaCO₃ nanoparticles loading, which made the surface structures more fragile. In terms of oxygen permeability, Nafchi et al. successfully enhanced the gas barrier properties of both gelatin and sago starch active packaging films by incorporating nanorod ZnO at 5 wt.% concentration [211]. However, ZnO nanoparticles can pose environmental hazards to aquatic life, as they cannot be easily assimilated in the ecosystem, raising some concern about eco-friendliness of such renewable biodegradability nanocomposite films.

4.4. Emulsion-based edible coatings and films

The addition of rationally designed layers of edible coatings with excellent moisture/oxygen barrier properties to prolong the shelf-life of fruits and other fresh produce is an appealing solution to mitigate spoilage concerns. Among all edible coatings, the emulsion-based coatings stand out as one of the most developed designs. These edible coatings and films are not meant to replace conventional packaging completely but rather provide secondary support for food protection by enhancing barrier properties for the primary food packaging materials [219]. Many other reviews that focus on the processes, antimicrobial, and antioxidant properties of the emulsion-based edible coatings and films [220,221]. Though, not many have reported on the improvements in moisture and gas barrier properties.

There are many micro and nano emulsion blends combinations to produce edible coatings and films. The barrier properties of the blend films must be governed by two main components: the hydrocolloid part, which provides good mechanical and oxygen properties, and the lipid phase, which gives rise to sound moisture barriers [130]. In most of these systems, the lipid (oil phase) will be emulsified with the aqueous hydrocolloid prior to casting films or fabricating an edible coating on goods. Because of their ultra-low oil-water interfacial tensions, micro and nanoemulsion coatings are typically homogeneous, isotropic, and thermodynamic stable colloidal systems. However, the stability of the emulsion blends and the barrier properties of the resulting films and coatings are governed by the droplet size and zeta potential charge of the lipid compounds [36]. Typically, smaller lipid droplet size and lower zeta potential charge would give better and extended emulsion stability to the coating formulation, which would require extensive high energy homogenization techniques such as the processes of microfluidization, ultrasound, and sonication [130,222].

Moreover, incorporating traditional surfactant or Pickering emulsifiers usually enhances the stability of emulsions [131]. For example, lecithin, a natural ingredient extracted from egg yolk or soy, is widely used as an emulsifier in the pharmaceutical and food

Table 6

Oxygen/water vapor improvements of edible emulsion-based coatings and their emulsion stabilizer for some of the food products.

Lipid compound	Hydrocolloid compound	Emulsion stabilizer	Product coatings	WVP/ WVTR	OP/ OTR	Ref
Classical emulsions						
Basil essential oil & Beeswax	Chitosan	Tween 80	Eggs	✓	-	[227]
Vegetal essential oil	Chitosan	Tween 80	PLA surface coating for beef meat and white cheese	-	✓	[228]
Aloe vera	Chitosan	Tween 80	Blueberries	✓	-	[230]
Egg yolk (fatty acid)	Egg white (poly(albumin)) reinforced with CNC	Egg yolk (lecithin)	Avocado, Bananas, Strawberries, Papayas	✓	✓	[224]
Pickering emulsions						
Beeswax	Cornstarch	CNC-dodecyl succinic anhydride (CNC-DDSA) modified nanoparticles	Apples, Bananas, Strawberries,	✓	✓	[231]
Beeswax	Carboxymethyl chitosan	CNF	-	✓	-	[232]
Cinnamon essential oil & Perilla essential oil	Collagen	Chitosan nanoparticles	Fish fillets	✓	-	[233]
Cinnamon essential oil & corn oil	Sodium alginate	Sodium starch octenyl succinate (OSA starch)	-	✓	✓	[234]
Clove essential oil	Chitosan	Zein colloid particles	-	✓	-	[235]
Oleic acid	Chitosan	CNC	Pears	✓	-	[236]

industry to stabilize emulsion due to its consumption safety and cost-effective attributes [223]. Jung et al. recently developed a multifunctional cellulose reinforced nanocomposite emulsion blend of egg-derived polymers poly(albumin) and egg yolk stabilized with lecithin, as shown in Fig 17 [224]. The coatings possess superior gas and moisture barrier properties compared to other natural-based coating such as starch, pectin, and carnauba waxes, and are comparable to high barrier materials such as PET (Fig. 17a). In addition, the coatings have been shown to reduce antimicrobial growth and slow down respiration rate, as well as to prevent dehydration and keep the fruits more polished for cosmetic purposes (Fig. 17b). Traditional surfactants such as Tween 80 and Tween 20, commonly used in foods and cosmetics, have also demonstrated decent emulsifying properties to stabilize essential oil droplets in film-forming emulsions [225–228]. However, the uses of these traditional surfactants in food coating applications are controlled at a strict level due to limited biocompatibility [131].

In recent years, Pickering emulsions obtained from natural resources are getting tractions as a favoured alternative method to stabilize emulsions. Pickering emulsions are stabilized by solid particles at oil/water interfaces, requiring less emulsifiers to stabilize [229]. Various compounds used in the fabrication of emulsified blend films and coatings and their respective emulsion stabilizers are listed in Table 6. These emulsion-based edible coatings possess good oxygen/moisture barrier properties, which can further extend the shelf-life of the foods by preventing the browning effect, oxidative rancidity, and moisture loss. It is also important to emphasize that these micro and nanoemulsion-based coatings and films can provide extra physicochemical, antioxidant, and antimicrobial properties to the product coatings.

4.5. Paper coatings

Paper is one of the most appealing materials extensively used in food packaging industries, mainly because of its biodegradability, lightweight, recyclability, and oil resistance. However, the utilization of paper is limited by its hydrophilicity and porosity, which ultimately inhibit its usage as protective coatings against the adsorption of water, gases, and oils. In addition, base papers typically display inferior barrier properties due to their largely porous structure, which would allow moisture and oxygen gas particles to pass through the membrane effortlessly. Thus, paper barrier resistance in packaging applications is traditionally governed by coatings derived from petroleum-based materials, such as PE and paraffin waxes [237]. Besides the moisture/oxygen barrier properties, the paper coatings were also designed to provide barrier protection against grease/oil, water, and other gases (CO_2/N_2) to prolong the shelf life of packed food, as well as the loss of flavors, aroma, and colors. However, the uses of these paper coatings have become less desirable due to poor recyclability and environmental concerns despite their advantages in inhibiting moisture/oxygen permeability.

Alternatively, many recent technological advances in paper coating focus on designing sustainable coatings utilizing natural macromolecules, including polysaccharides, proteins, lipids, lignin, bio-polyesters and their derivatives [238–241]. Some investigations produced designs and formulations that provide fully biodegradable paper coatings while maintaining protection against

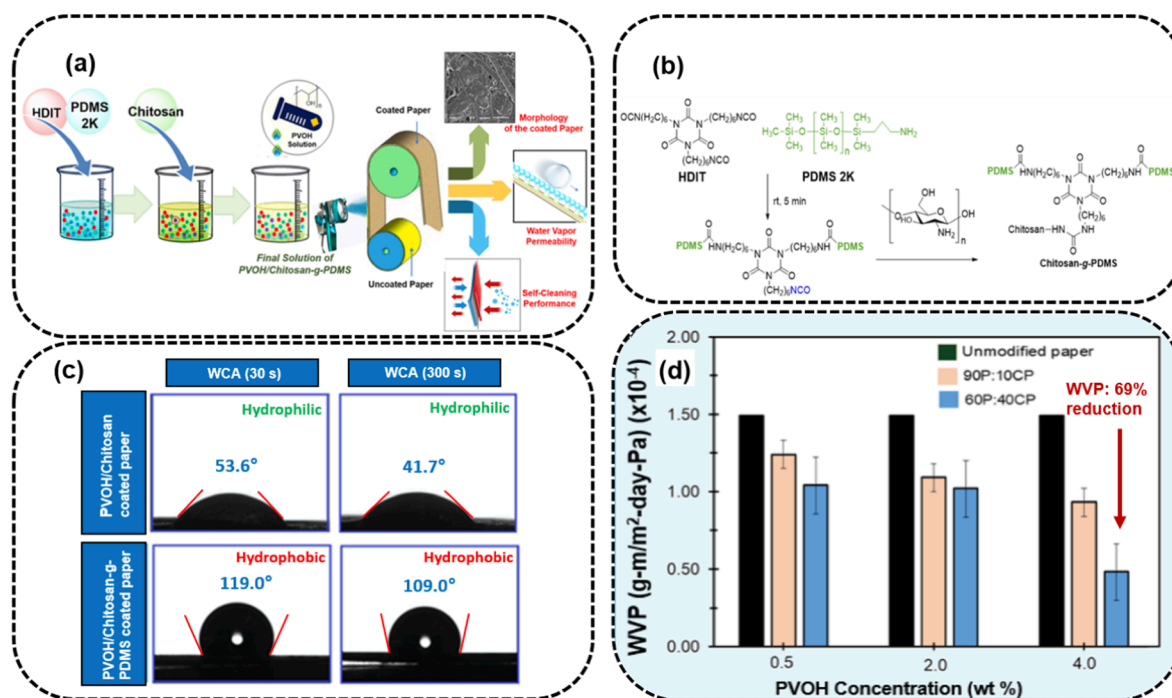


Fig. 18. (a) Processing of hydrophobic Chitosan-graft-PDMS and PVOH solution coating for Kraft-liner paper. (b) Reaction scheme to synthesize Chitosan-g-PDMS. (c) Hydrophobicity improved with the grafting of PDMS, indicated by water contact angle (WCA) results. (d) Reduction in WVP values of the hydrophobic blend coating. Adapted with permission from ref. [185] from Wiley Online Library. Copyright ©2021.

Table 7

Oxygen/water vapor improvements of multiphase paper coatings compared to base papers.

	Substrate	Biopolymer coating	Coating process	Coat grammage (g/m ²)	Coat thickness (μm)	OTR/OP improvement (%)	WVTR/ WVP improvement (%)	Ref
Blend	Kraft-liner paper	Chitosan-g-PDMS & PVOH	Spray coating	2.79	15.5	-	69% (23°C, 50% RH)	[242]
		CMC & AgNP modified CNC	Dip coating	-	187	45.4% (22°C, 48% RH)	93.3% (22°C, 75% RH)	[246]
	Paperboard Wrapping paper	CMF & CNC	Rod coating	7.8	260	-	25% (23°C, 50% RH)	[247]
		Alginate & CMC & Carrageenan & Grape seed extract	Rod coating	7.6	11	-	29% (23°C, 50% RH)	[248]
	Manila paper	PBAT & palmitoyl chloride modified lignin	Rod coating	14.2	29	55.2%(22°C, 48% RH)	-	[249]
	Multilayer assembly	PLA & Nanocellulose multilayer	Extrusion & slot-die coating	-PLA layer: 23.4-Nanocellulose layer: 6.7	-PLA layer: 17.7-Nanocellulose layer: 4.3	Improved* (N/A)	92.3% (23 °C, 50% RH)	[243]
		Carrageenan & Chitosan & MMT multilayer	Layer-by-layer	80	-	90% (23 °C, 50% RH)	81% (37 °C, 75% RH)	[250]
		Nanocellulose-crosslinked with Citric acid	Layer-by-layer dip coating	61	30	99.5% (22°C, 49% RH)	87% (22°C, 60% RH)	[251]
	Micro and nanocomposites	Kraft paper	Rod coating	21.99	29	-	78% (23°C, 50% RH)	[252]
		Whatman filter paper	Spray coating & corona discharge	43	29	Improved* (N/A)	86% (25°C, 50% RH)	[244]
		Kraft paper	Rod coating	65	-	-	26% (23°C, 50% RH)	[253]
		Manila paper	Blade coating	-	100	92% (23 °C, 30% RH)	87% (23 °C, 50% RH)	[254]
		Kraft paper	Rod coating	20.2	107.7	100% ** (23 °C, 50% RH)	82.6% (23 °C, 50% RH)	[245]

* Data for base paper was not obtained due to the OP values for papers were out of upper range.

** Data OP values of coatings were dropped by a factor of thousands or millions

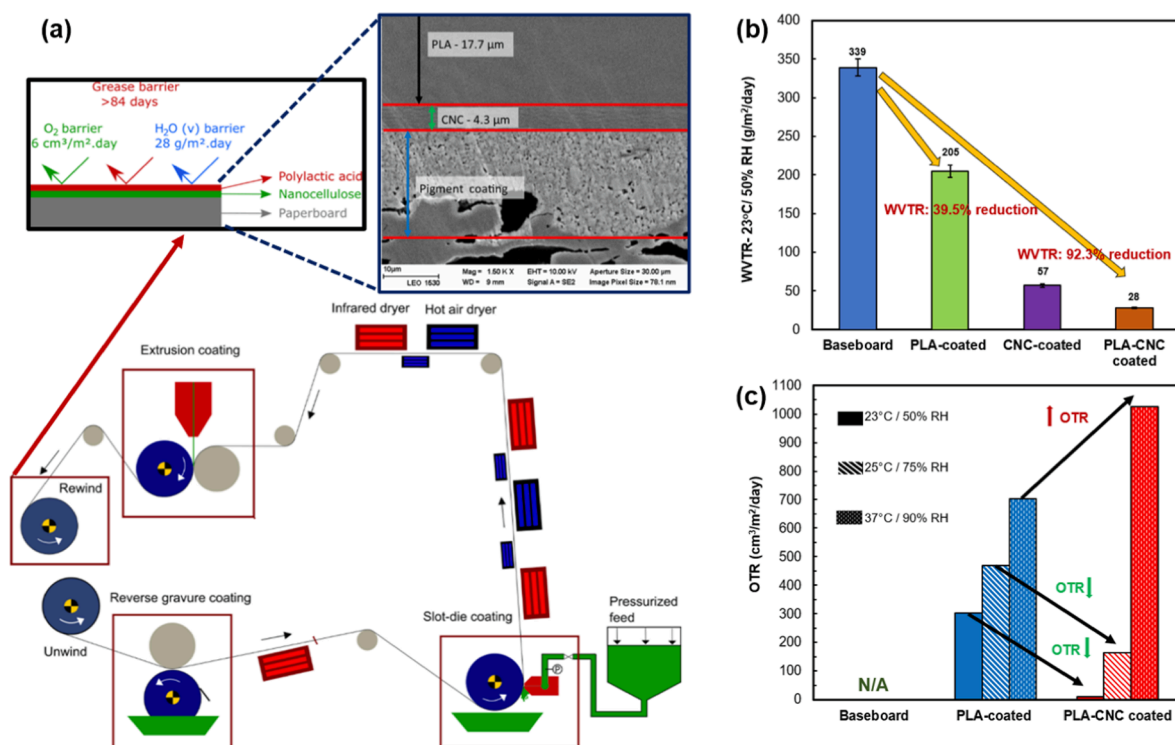


Fig. 19. (a) Scheme for the slot-die and extrusion fabrication process of PLA-CNC bilayer paper coatings. (b) Improved WVTR and (c) OTR of PLA-CNC bilayer coatings on paperboard. Adapted with permission from ref. [243] from American Chemical Society. Copyright ©2019.

moisture/oxygen permeation. These approaches include blending multiple biopolymers, incorporating nanoparticles in coating formulations, surface modifications, and utilizing compatibilizers [237]. For instance, Nair et al. fabricated a blend of PVOH and modified chitosan-graft-poly(dimethylsiloxane) (PDMS) as a hydrophobic coating for Kraft-liner papers (Fig. 18) [242]. The authors found that by grafting PDMS onto chitosan using surface linking agent hexamethylene diisocyanate trimer (HDIT), the affinity of the biopolymer changed from hydrophilic to hydrophobic noted from water contact angle (WCA) measurement. As a result, the modified blend coatings showed considerable improvements in moisture barrier substantiated by a 62% and a 69% improvement from water Cobb (Cobb60) and water vapor transmission rate (WVTR) test.

The barrier improvements of some exemplary biodegradable paper and paperboard coatings are summarized in Table 7. The paper coatings presented in Table 7 are fabricated using various lab-scale techniques, including spray coatings, rod coatings, blade coating, and dip coatings. Koppolu et al. introduced a more scalable fabrication method of bio-based and biodegradable paper coatings using a slot-die and extrusion coating (Fig. 19) [243]. The authors successfully produced a nanocellulose and PLA bilayer coatings that significantly inhibit moisture oxygen permeation through this process. More specifically, this bilayer coating displayed 92.3% WVTR reduction compared to baseboard and 98% OTR reduction compared to the PLA monolayer coated baseboard.

Though paper coatings are reported to immensely improve the oxygen barrier properties of non-coated paper substrates, the oxygen permeability values of base papers are still inferior and usually out of range using various oxygen permeability measurement techniques. Thus, estimations of oxygen barrier improvement compared to base papers cannot be directly obtained, as evidenced by multiple studies, such as those reported by Koppolu et al. [243] and Mirmehdi et al. [244]. However, it can be expected that the oxygen permeability can be reduced by a factor of thousands or millions. A recent study by Zhang et al. on the antibacterial composite paper coatings made from acetylated cellulose and cinnamaldehyde (CIN) showed that the oxygen permeability was reduced by a million times, and the overall oxygen barrier properties were improved essentially by 100% [245]. The results from their meat packaging application suggested that the addition of high barrier and antibacterial coatings on paper wrappings can effectively extend the shelf-life of beef in the fridge by additional 4-5 days.

4.6. Other technologies

Besides blocking the permeation of water vapor/gas molecules through film substrates by high barrier polymers or fillers, other stratagems, such as employing oxygen (and gas) scavengers, moisture absorbers, and metallization to further improve the barrier properties of multicomponent sustainable polymer systems are reported in the literature and employed in some commercial applications (Fig. 20).

Some active and smart packaging designs encapsulate gas (O₂, CO₂, or C₂H₄) scavengers in the packaging material to absorb

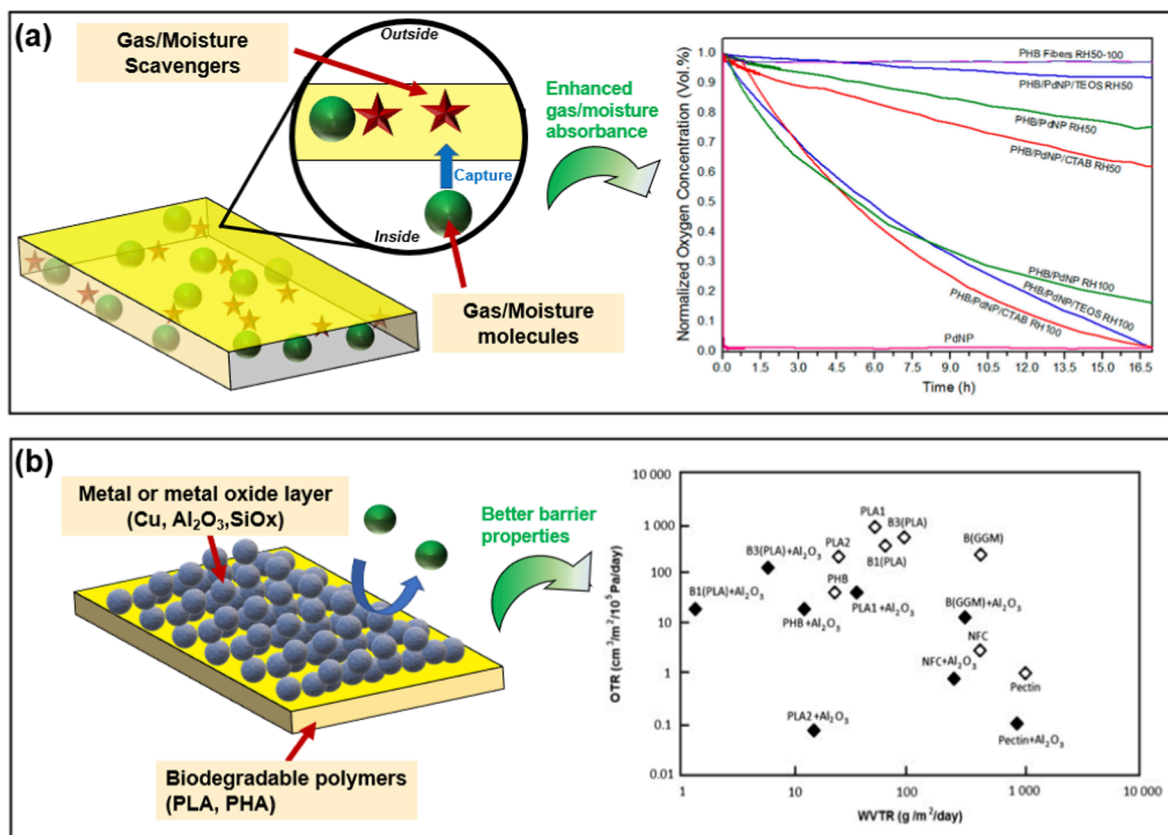


Fig. 20. Illustration of (a) Gas/Moisture scavenger technologies in biodegradable polymer systems and their influence on absorbance capacities. Adapted with permission from ref. [257] from MDPI Open Access. (b) Metallization technologies in biodegradable polymer systems and the comparison on the overall barrier properties of commercialized bioplastics. Adapted with permission from ref. [279] from American Chemical Society. Copyright ©2011.

permeating gas or dissolved gas in the packaging headspace to diminish the contact time of food with these gases [5,255]. The incorporation of these gas scavengers slow down deterioration, prolonging shelf-life, and maintain the effective quality of food products [256]. Typical oxygen scavengers that have been used in sustainable packaging films are iron sachets, α -tocopherol, ascorbic acid, and palladium [256–259]. Like barrier properties, the gas scavenging performance of the packaging material is dependant on the dispersion of the scavenging material in the film matrices. Cherpinski et al. reported a clear improvement in oxygen scavenging activities of PHB/Palladium nanoparticle (PdNP) composites when hexadecyltrimethylammonium bromide (CTAB) was added in the formulation as dispersing aids [257]. Activated carbon (AC) has also been widely used in active packaging as CO₂ [260,261] as well as ethylene gas scavenger (C₂H₄) [262]. The high surface area, extensive network, and porosity presented in AC provides outstanding CO₂ scavenging capability [263,264]. Moreover, AC can also be utilized for their excellent adsorption capabilities and function as a containing medium for other oxygen/gas/moisture scavengers [265,266] or antimicrobial agents [267,268]. For instance, Joven et al. developed an oxygen scavenging composites by impregnating sodium erythorbate (SE) inside the channels networks of powdered activated carbon (PAC) as fillers of packaging films, which promote the capturing and trapping of O₂ in PAC particles [266].

Similar to oxygen scavengers, water vapor/moisture absorbers and desiccants are also employed as effective tools to minimize the humidity and water content of food, which help reducing microbial contamination and stalling the overall spoilage rate of goods [269,270]. In general, moisture absorbers should be hygroscopic so that they absorb and retain water from the packaging headspace environments. Thus, moisture scavengers can either exist as side desiccant sachets/pads or imbedded in the polymeric films as those are the two most effective form to absorb moisture and control humidity inside the packaging spaces [270]. Example of natural-based moisture absorbers that have been extensively researched include sorbitol, fructose, and carboxymethyl cellulose. However, the moisture absorption capacity of these natural moisture scavengers are still relatively lower than inorganic material-based desiccants, such as silica gel [271], calcium chloride [272], potassium chloride [273], bentonite [274], and attapulgite poly(acrylamide) [275]. Nevertheless, it has been shown that a combination of different moisture absorbers could achieve enhanced humidity control. Mahajan et al. found that the combination of fast absorbing scavengers (sorbitol and calcium chloride) and slow moisture scavengers (bentonite) provided the best moisture absorbing and holding capacity for about 5–6 days, while pure scavengers have either very slow absorption or poor moisture holding performances [276].

Metallization is yet another very effective method to generate biopolymer-based packaging materials with superior barrier

properties, in which impermeable thin layers of metals are deposited on the surface of the polymer film substrates [277,278]. Barrier properties provided by metallization are subjected to influence by the coating thickness and surface treatment of polymer substrates [279,280]. Due to the excellent water vapour/gas barrier properties that these metal layers possess, metallization is commonly utilized as modified atmosphere packaging (MAP) in the medical or food packaging industries to make up for the poor barrier capabilities of biodegradable polymers [277,281]. Romani et al. compared the permeation of metallized PP and metallized PLA and found out that there is no significant difference in terms of water/gas barrier despite PLA being more permeable to gases, indicating that most of the barrier properties are contributed by metal layer [282]. Common metals including copper (Cu) [283], aluminum oxide (Al_2O_3) [279,284,285], or silicon oxide (SiO_x) [281,286,287] have been deposited on the surface of PLA by either physical vapour deposition (PVD), chemical vapour deposition (CVD), or atomic vapour deposition (AVD) techniques. The WVP and OP values of these metallized PLA composites have reduced by 600 – 1000%, compared to uncoated PLA, indicating the low permeation rate of metallized biopolymer composites. Metallization and their environmental performance of other biodegradable polymers, such as PHAs were also investigated. For example, a recent work by Vea et al. evaluated the lamination of PHA with Al or AlO_x to promote barrier properties [288]. Similar to the metallization of PLA, the authors observed that the employment of Al or AlO_x provided outstanding barrier properties, though, other aspect such as economic and socially oriented analyses need to be considered to fully implement metallization of PHA with these metal layers.

There is a considerable concern on the sustainability implications of metallization of compostable polymers, such as PLA and PHAs as it may stagnate or compromise their compostability or biodegradability [289]. Most of the researches on the metallization of biopolymers have all acknowledged the shortcoming in term of biodegradability as these metal layers are non-biodegradable and may cause challenges in the recycling of the barrier packaging materials [279,280,286]. Thus, it has been suggested that metallization should only be used as a temporary solution to provide barrier properties, and more extensive research should focus on other technologies that utilize biodegradable materials through layer-by-layer (LBL) assembly [45].

5. Sustainability of renewable multicomponent packaging

Sustainability is one of the most important focal points to put into evaluation when developing these classes of green packaging materials. The biodegradability of biopolymers used for barrier packaging should follow metrics and methods set out by different standards, depending on which compositing or degradation processes that the biopolymers are subjected to. First-generation or nature derived biopolymers, such as cellulose, starch and lignin, are biodegradable in various environments, including home and industrial composting, soil, fresh water, aerobic digestion, and the landfill (Fig. 21). Chitosan, protein, alginate, and other bioplastics, such as PLA and PHA, are other attractive materials used for fabrication of sustainable packaging films. Among these materials, PHAs, starch and cellulose have excellent biodegradability in both landfill and marine environments, providing more advantages for various applications (Fig. 21) [290]. On the other hand, polyesters, that are considered as sustainable bioplastics, such as PBAT (typically petroleum derived), PBS (can be bio-based or petroleum derived) display good biodegradability in only industrial composting testing standards (Fig. 21). It is also important to consider the biodegradability of compatibilizers, fillers, and additives as they are parts of multi-component blends or composites. Currently, many of the renewable packaging systems share the same petrochemical-derived and non-biodegradable additives that are traditionally used in non-biodegradable petroleum-based packaging [291,292]. The utilization of bio-based and biodegradable alternatives can be a suitable option to ensure the complete biodegradation or compostability of sustainable polymer systems [293].

Alternatively, recycling and reprocessing of packaging materials is also a good option to consider in some situations, especially when non-biodegradable fillers and additives are involved [294]. The review by Gioia et al. discusses the merits and shortcomings of both composting and re/upcycling methods that are applicable to biodegradable plastics [295]. Currently, most materials used for packaging, such as aluminum, paper, paperboard, plastic, and woods, all undergo separation and recycling system that are well-established to re-utilize the materials. For biopolymers, reusing and recycling hold little merits, and in some cases severe limitations, most significantly the loss of mechanical and barrier performances due to thermal degradation and chain scissions after a few rounds of reprocessing.

There have been numerous investigations on the sorting and recycling of PLA, as it is regarded as the premier biopolymers to be used in barrier packaging [296]. It was reported that the presence of PLA impurities in other packaging materials can greatly influence the rheology of the recovered material and reduce the mechanical and barrier performances [297]. Issues arising from the contamination from PLA incentivize better efforts to develop a working technology that can separate PLA from multicomponent wastes. Sánchez and Collinson investigated on the methods of separating PLA from PLA/PET blends for water bottle by depolymerizing PLA into oligomer liquid using zinc acetate and filter PLA from the mixture [298]. This is an effective strategy to separate PLA and PET, as PLA/PET blends cannot be separated by using conventional physical separation method of density sorting because of their similar density [299]. However, the recycling of PLA can still be problematic since it would require hefty cost investments for a technologically viable system, mainly because the separation process for PLA is not yet well-established. Regarding the separation and cleaning process of other biodegradable/compostable polymers, such as PBAT, PBS, PCL, PHA, their blends, and composites, the current state of knowledge is still limited compared to PLA-based multiphase materials [295]. Thus, further research and investigation on the recyclability of these polymer systems is required.

6. Challenges, future trends, and concluding remarks

As society is shifting towards a more sustainable lifestyle, there is a dire need to substitute single-use plastics in our day-to-day

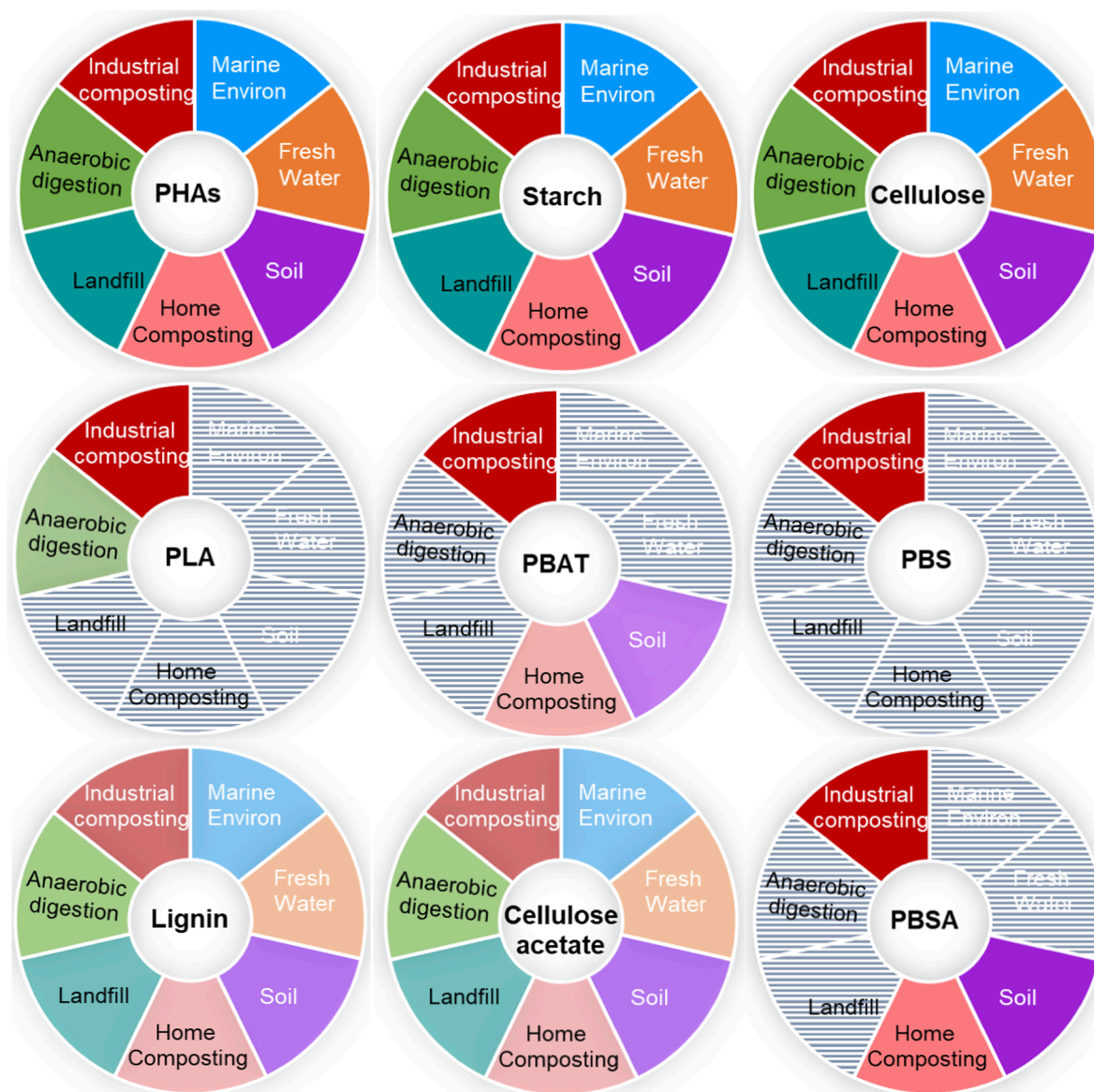


Fig. 21. Comparison of biodegradation attributes of biopolymers in various environments*. Bold colors indicate proven biodegradability; lighter colors indicate biodegradability under certain conditions or for certain grades, striped – lines indicate non-proven biodegradability. *Environments: industrial composting (58 °C, 90 % biodegradation; 6 months); Aerobic digestion (Thermophilic 52 °C; mesophilic 37 °C; 90% biodegradation); Landfill (no standard specifications); Home composting (28 °C, 90% biodegradation, 12 months); Soil (25 °C, 90% biodegradation, 2 years); Fresh water (21 °C, 90% biodegradation, 56 days); marine environment (30 °C, 90% biodegradation, 6 months). Reproduced with permission from ref [300].

activities. Packaging materials used in consumer foods constitute a substantial portion of the single-use plastic market. Thus, new technologies and innovations in sustainable polymers should account for performance parameters required in food packaging and coating applications, extending to pharma and biomedical packaging. While there has been considerable progress in sustainable polymers over the last two decades, their use in barrier packaging is limited due to inferior moisture/gas resistance and high cost compared to the incumbent polyolefin-dominated materials, such as PET, PE, PP, and EVOH. As individual materials, most biopolymers in the current market often possess either moisture or oxygen barrier properties but rarely have both. However, the rational design of multiphase and multicomponent systems can improve the barrier properties. As such, innovations to fabricate multiphase and multicomponent systems, such as blending, multilayer assembly, or fabrication of nanocomposites, need to be adopted to overcome high moisture/oxygen permeability limitations.

This review highlighted more than two decades of discoveries published by several research teams on factors influencing the moisture/gas barrier properties of biodegradable films and coatings, including chemical structure, polarity, crystallinity, chain

orientation, morphologies. Furthermore, technological innovations that enhance barrier properties, including blending and their compatibilization, multilayer assemblies, and nanocomposite fabrications, are represented. Mathematical models that can predict permeabilities in multicomponent polymer systems with a focus on packaging films and coatings are also presented.

Although there have been various successes in improving the moisture/oxygen barrier of multicomponent biopolymers and bioplastics, some challenges are prevalent and need to be addressed regarding the use of these multicomponent systems. Production cost poses some significant concerns as several fabrications method to create high barrier films, and coatings need specialized processes that are not integrated into the current polymer processing systems. While some sustainable polymers, such as PGA, have excellent barrier properties, their extensive use is limited by the high raw material cost. Moreover, for numerous biodegradable multicomponent systems, the polymeric constituents do not blend well, and compatibilizers are required to increase the interfacial interactions. However, there are limited studies that focus on incorporating compatibilizers to enhance the barrier properties of bioplastic blends and multilayers.

Nevertheless, a rise in the development and innovation of biodegradable blends and multilayers films will undoubtedly encourage more research to explore the use of both *in-situ* and *ex-situ* compatibilization approaches to generate high barrier packaging materials. Last but not least, multicomponent polymers currently used in the food industry still relies on high barrier non-biodegradable component, such as PET and EVOH, which only make the films and coatings partially biodegradable and not suitable for industrial composting processes. Therefore, developments on 100% biodegradable systems are of a high priority to produce materials that can be easily composted in a reasonable time frame as well as to avoid microplastic generation from partial degradation.

In the foreseeable future, biodegradable and sustainable aspects will be extensively utilized in food and pharmaceutical packaging due to their biocompatibility advantages over the current single-use plastics. Therefore, the high barrier properties of biodegradable film and coating are one of the core characteristics that will need to be considered when designing sustainable food packaging materials. Currently, the trends of combining multiple designs (blends, multilayer, nanocomposites) to obtain high barrier functions without compromising the cost are expected to attract considerable attention in polymer science and engineering. It is also important to highlight that multilayer packaging materials are complex materials composed of different layers of polymers that can be chemically dissimilar. Adding to their complexity, multicomponent polymer systems typically contain tie or adhesive layers, compatibilizers, and other additives. Thus, traditional plastic sorting and recycling technologies, such as gravity-mediated separation or thermomechanical reprocessing, cannot be directly applied to consider recycling as an end-of-life option or to recover value after a single use. As a result, any recycling effort must first involve a costly deconstruction process to separate the various layers. However, no commercially viable technologies currently exist to conduct such deconstruction, apart from selective solvent dissolution processes currently under investigation. Therefore, more innovations on robust moisture/gas barrier performance designs are required to achieve sustainable food packaging coatings and films that compete and outperform the incumbent traditional non-biodegradable plastics in the market.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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